

Lecture 12: Data Visualization Continued

Heather Mattie, PhD

October 8, 2026

Recipe of the Day!

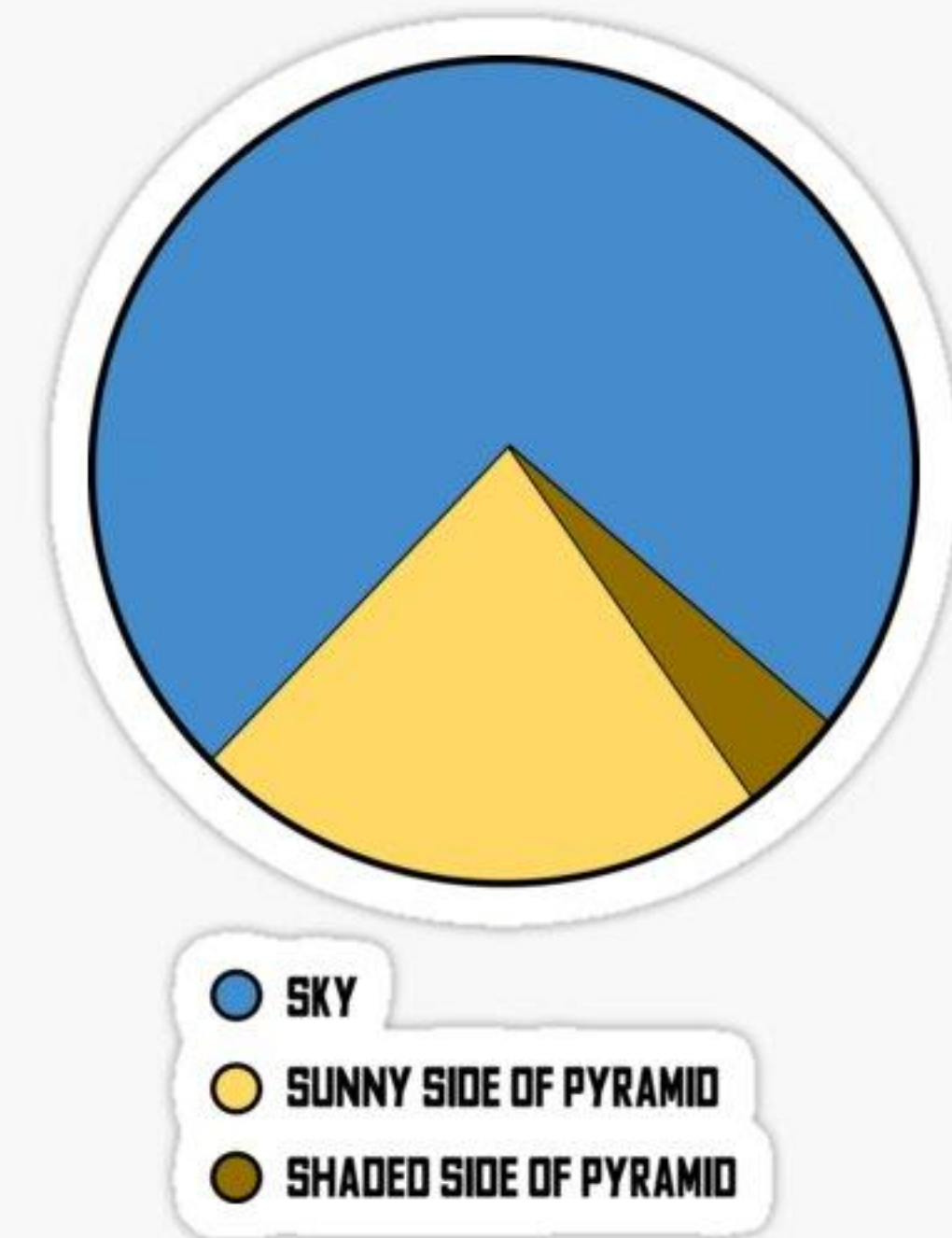
Epic Charcuterie Board



~ when you realize tomorrow is
Friday and Monday is a holiday ~

Announcements

- Heather's office hour: Thursday 12-1pm
- Homework #2 due Friday, October 9 by 11:59pm
- Lab will be held this week



Coding question of the day

Using the **murders** dataset, create a plot of murder rates with 4 boxplots next to each other - one boxplot for each region.

Order the boxplots by median rate (you will need the **reorder** function for this).

Make sure the boxplots are different colors, the axes are informative, and you add a title.

Remove the legend by adding this layer to your plot: **guides(fill = "none")**

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data("murders")
```

Comparing distributions

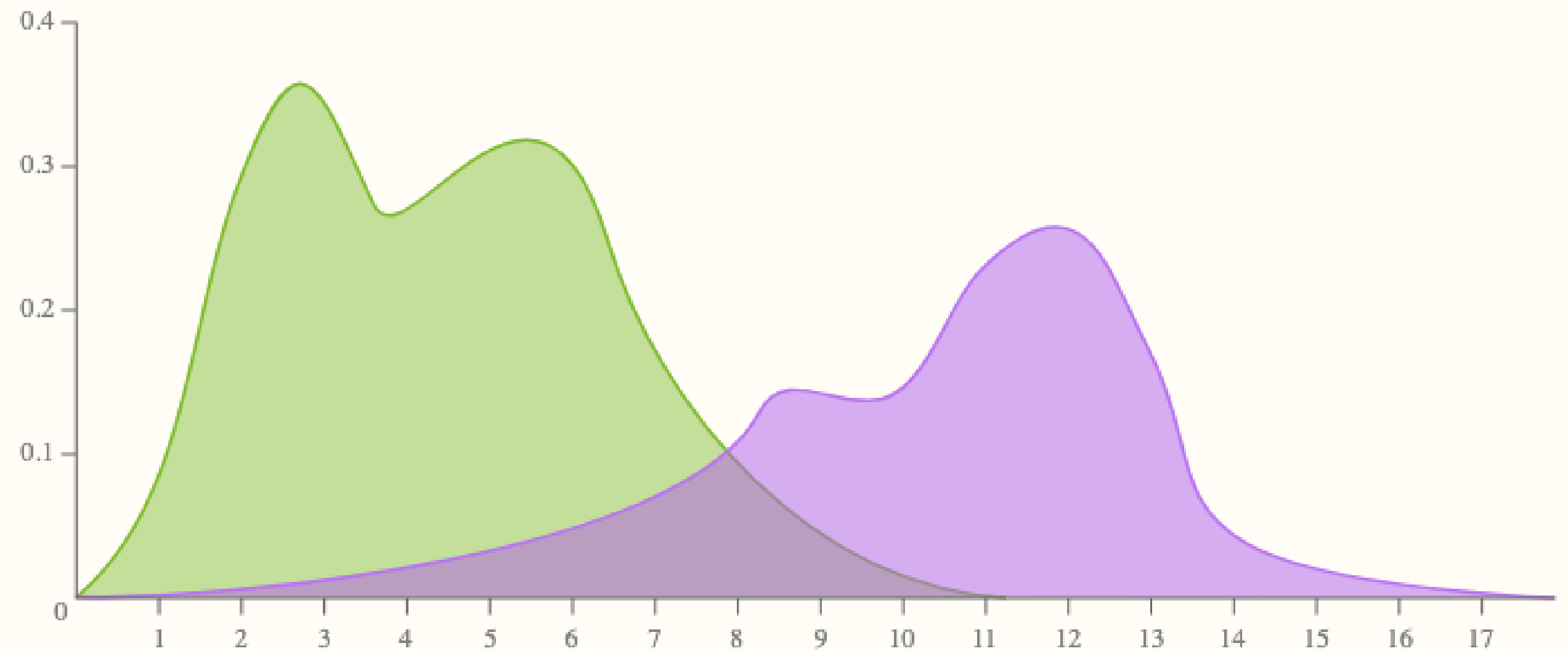
Smooth density plots

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Smooth density plots

Smooth density plots estimate the probability distribution of a continuous variable

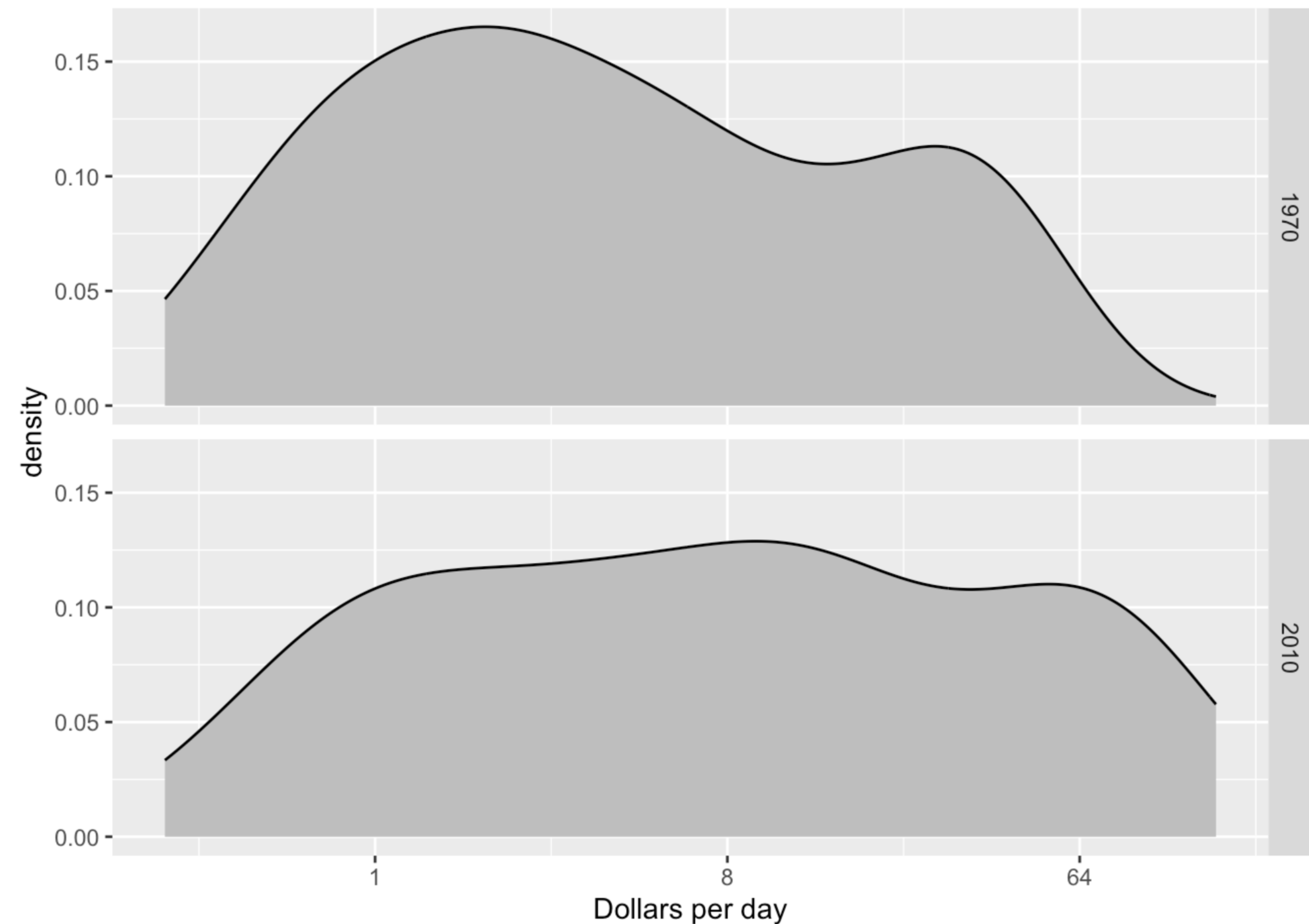
- A smoothed version of a histogram
- Useful for visualizing distribution shape
- Useful for comparing groups



Income distributions

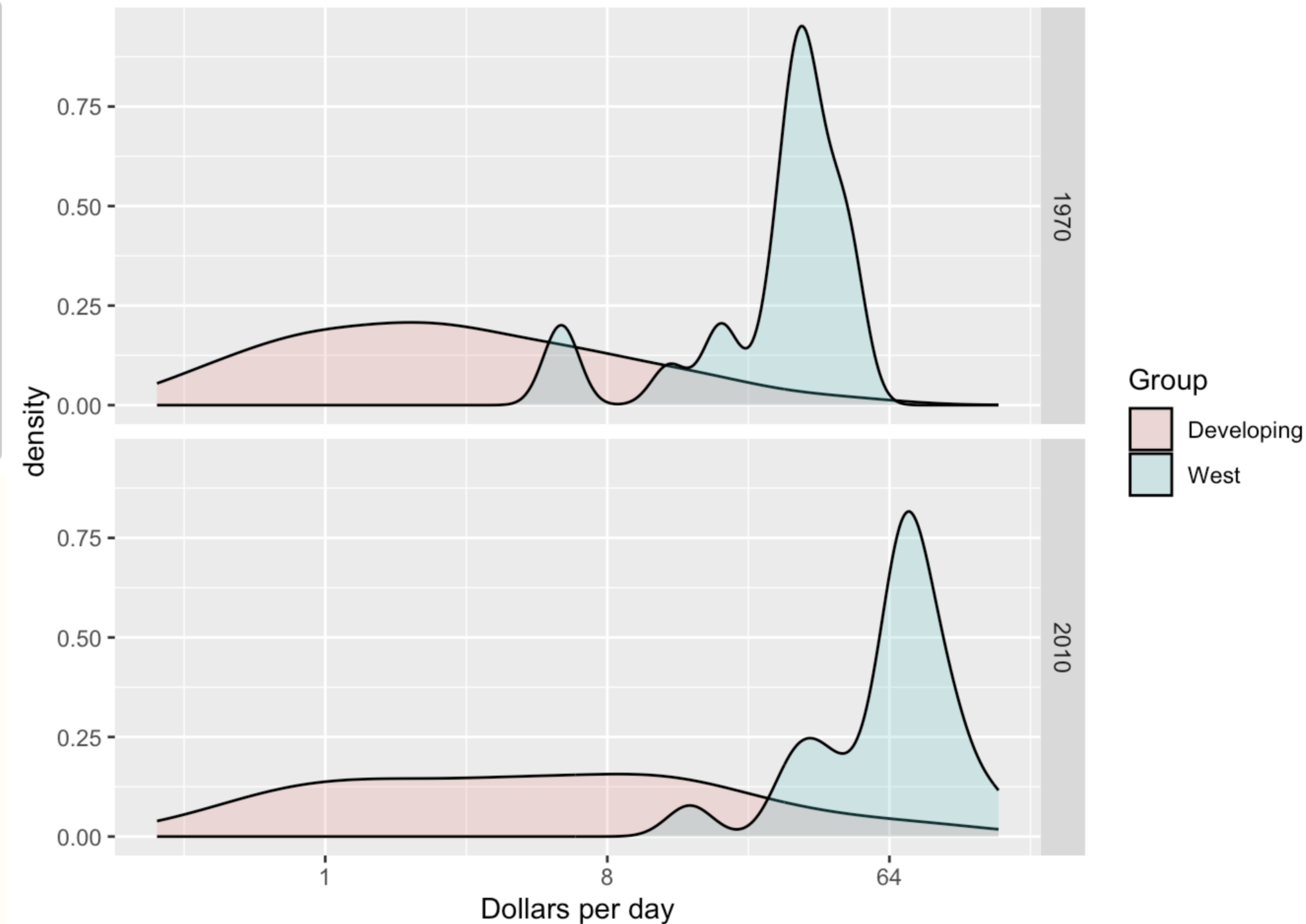
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
          country %in% country_list) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_density(fill = "grey") +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  facet_grid(year~.)
```

With this plot we can see that some countries have shifted to the right, closing the income gap – but we don't know **which countries** these are



Groups

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

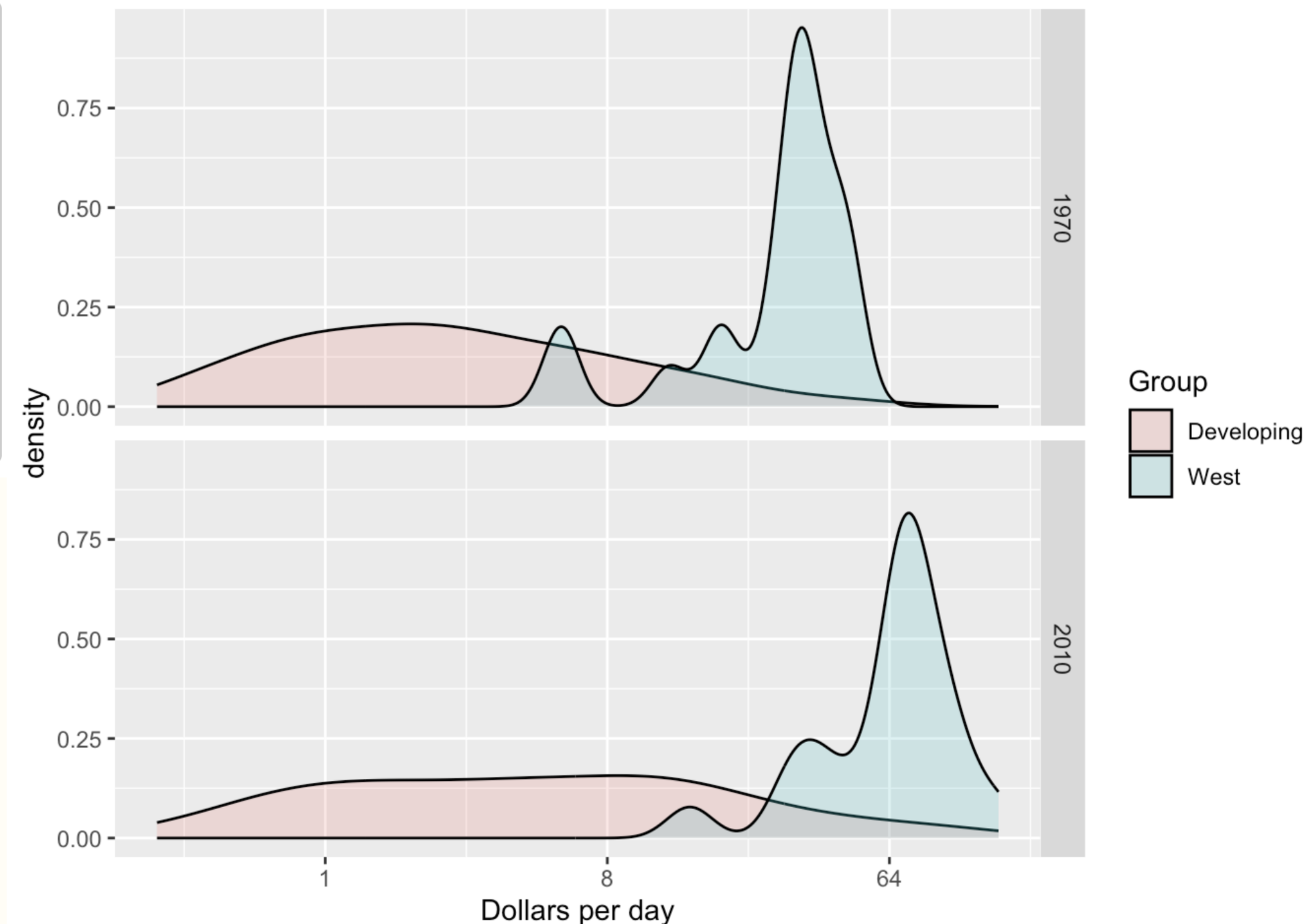


Groups

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

The default is for the area under a smooth density plot to be **equal to 1**

This is not ideal in this context since there are 21 countries in our West group and 87 in our Developing group

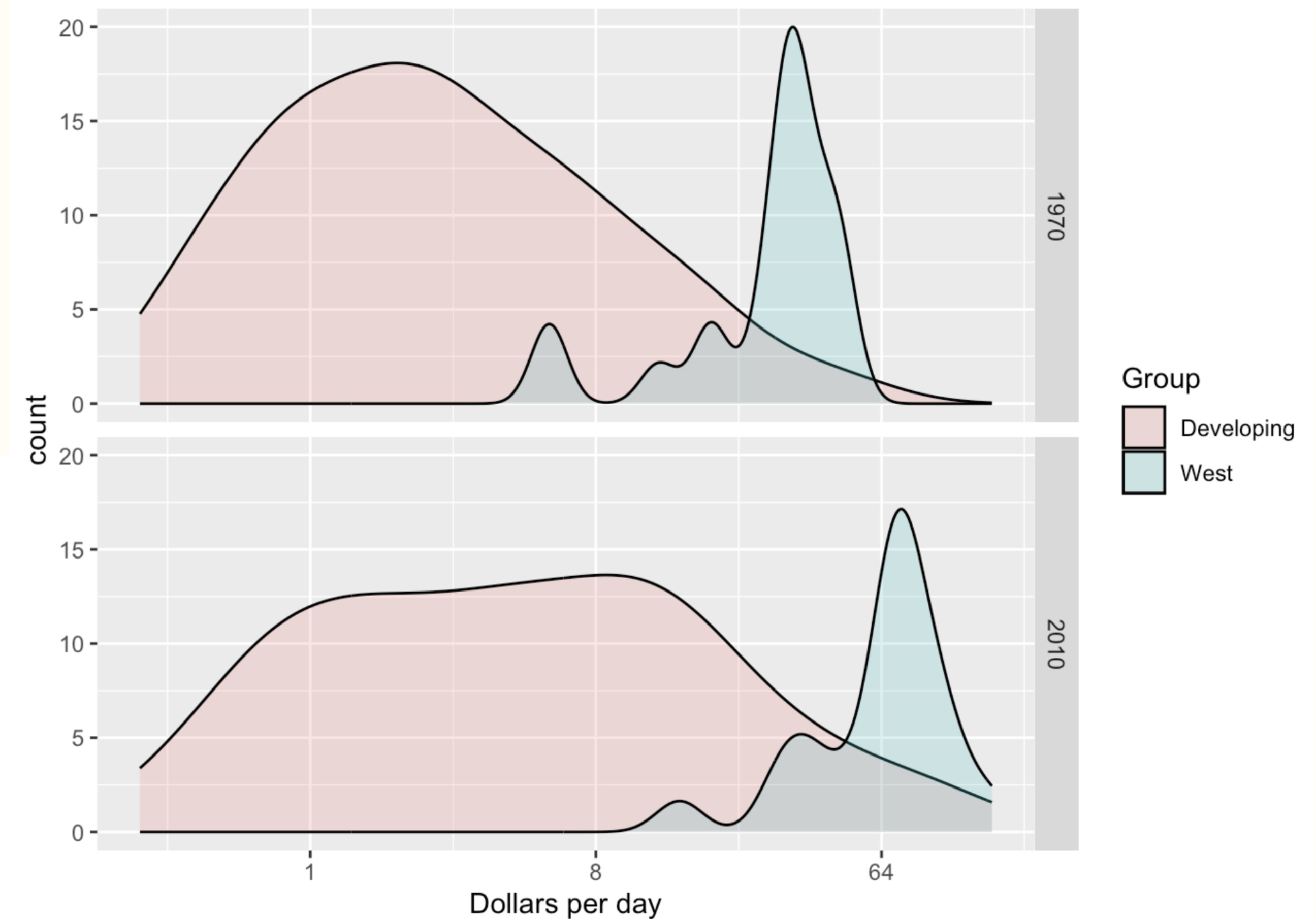


Accessing computed variables

Multiplying the y-axis values by the size of a group makes the area proportional to the number of countries in that group

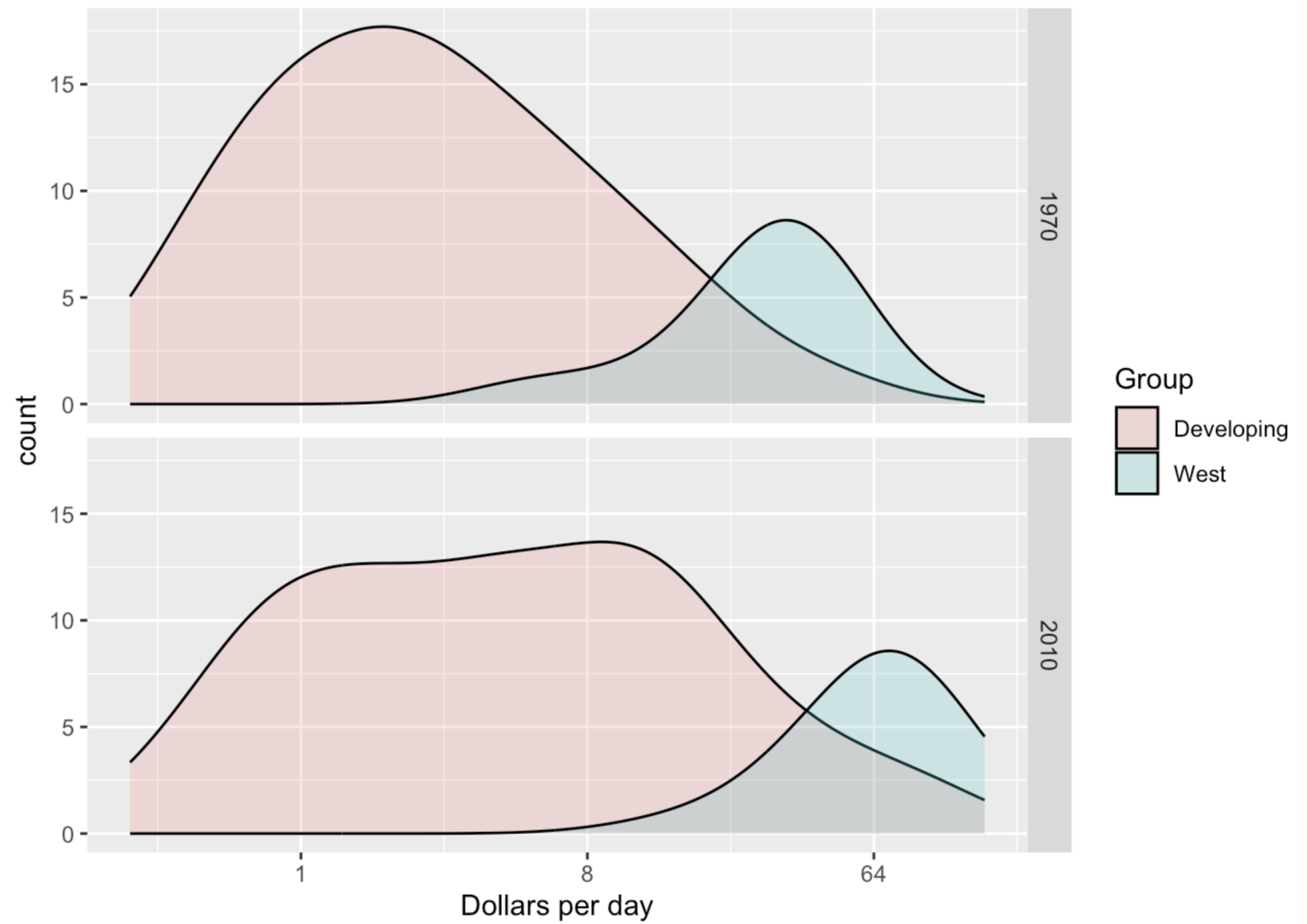
- **y = after_stat(count)**

```
gapminder %>%
  filter(year %in% c(past_year, present_year) &
         country %in% country_list) %>%
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +
  geom_density(alpha = 0.2) +
  facet_grid(year ~ .) +
  scale_x_continuous(trans = "log2") +
  xlab("Dollars per day") +
  labs(fill = "Group")
```



Smoother densities (**bw**)

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  mutate(group = ifelse(region %in% west, "West", "Developing")) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count), fill = group)) +  
  geom_density(alpha = 0.2, bw = 0.75) +  
  facet_grid(year ~ .) +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



case_when

We can make this plot more informative by having more group categories

- The **case_when** function can create a new group variable efficiently

```
gapminder <- gapminder %>%  
  mutate(group = case_when(  
    region %in% west ~ "West",  
    region %in% c("Eastern Asia", "South-Eastern Asia") ~ "East Asia",  
    region %in% c("Caribbean", "Central America", "South America") ~ "Latin America",  
    continent == "Africa" & region != "Northern Africa" ~ "Sub-Saharan Africa",  
    TRUE ~ "Others"))
```

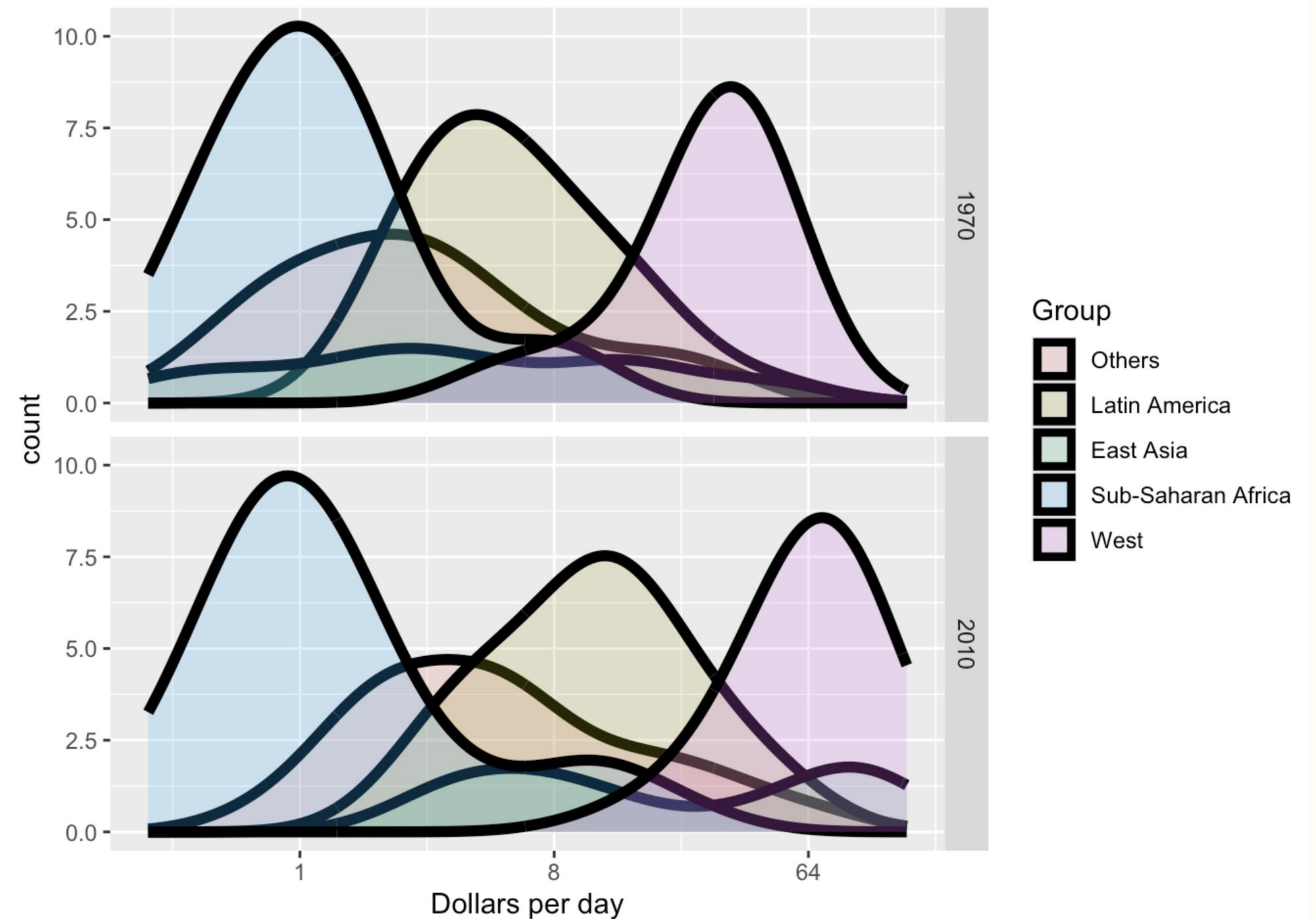
```
gapminder %>% select(region, group) %>% head()
```

##	region	group
## 1	Southern Europe	West
## 2	Northern Africa	Others
## 3	Middle Africa	Sub-Saharan Africa
## 4	Caribbean	Latin America
## 5	South America	Latin America
## 6	Western Asia	Others

linewidth

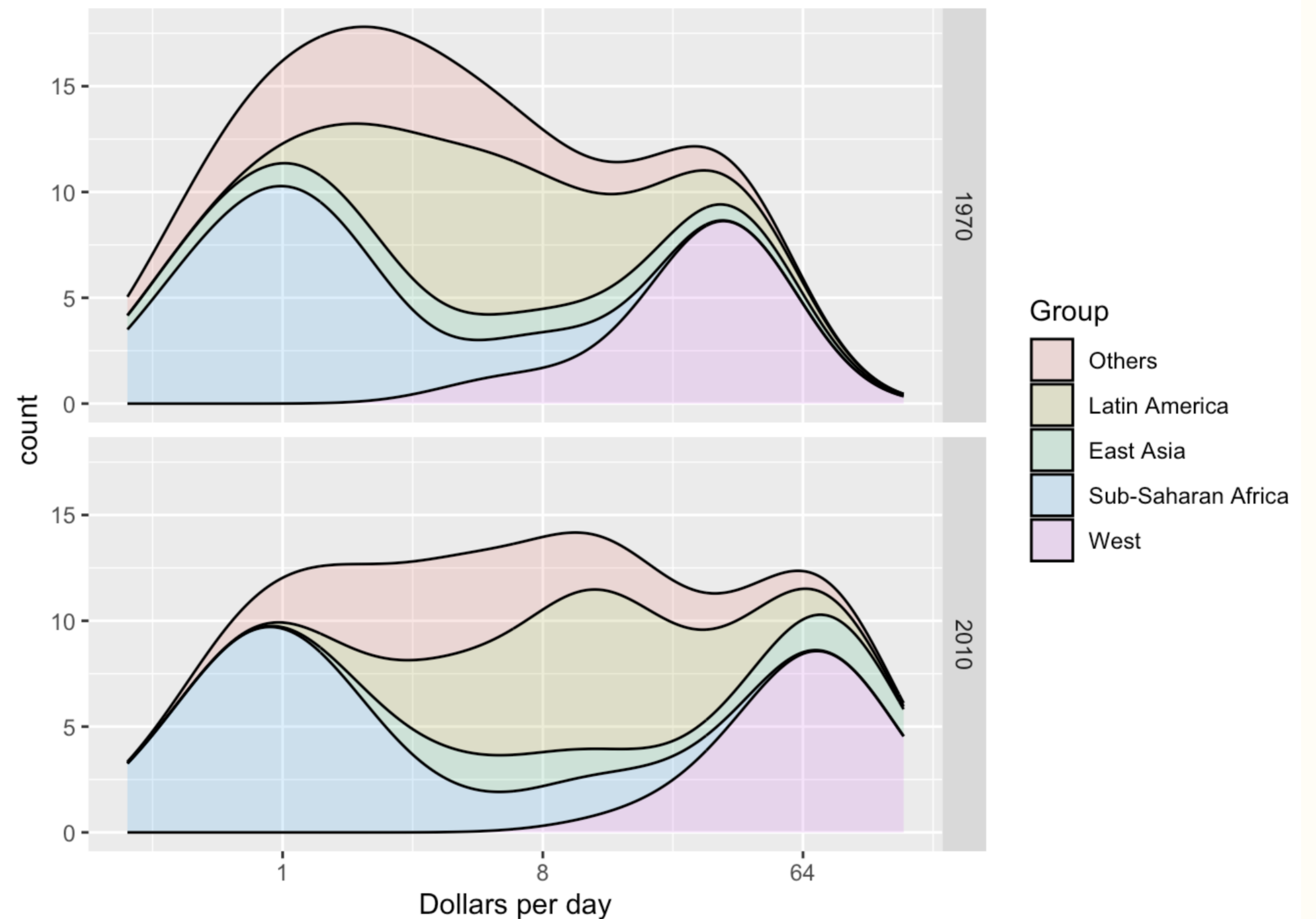
```
gapminder <- gapminder %>%  
  mutate(group = factor(group,  
    levels = c("Others", "Latin America",  
              "East Asia", "Sub-Saharan Africa",  
              "West")))
```

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
    country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
    fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, linewidth = 2) +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



position = "stack"

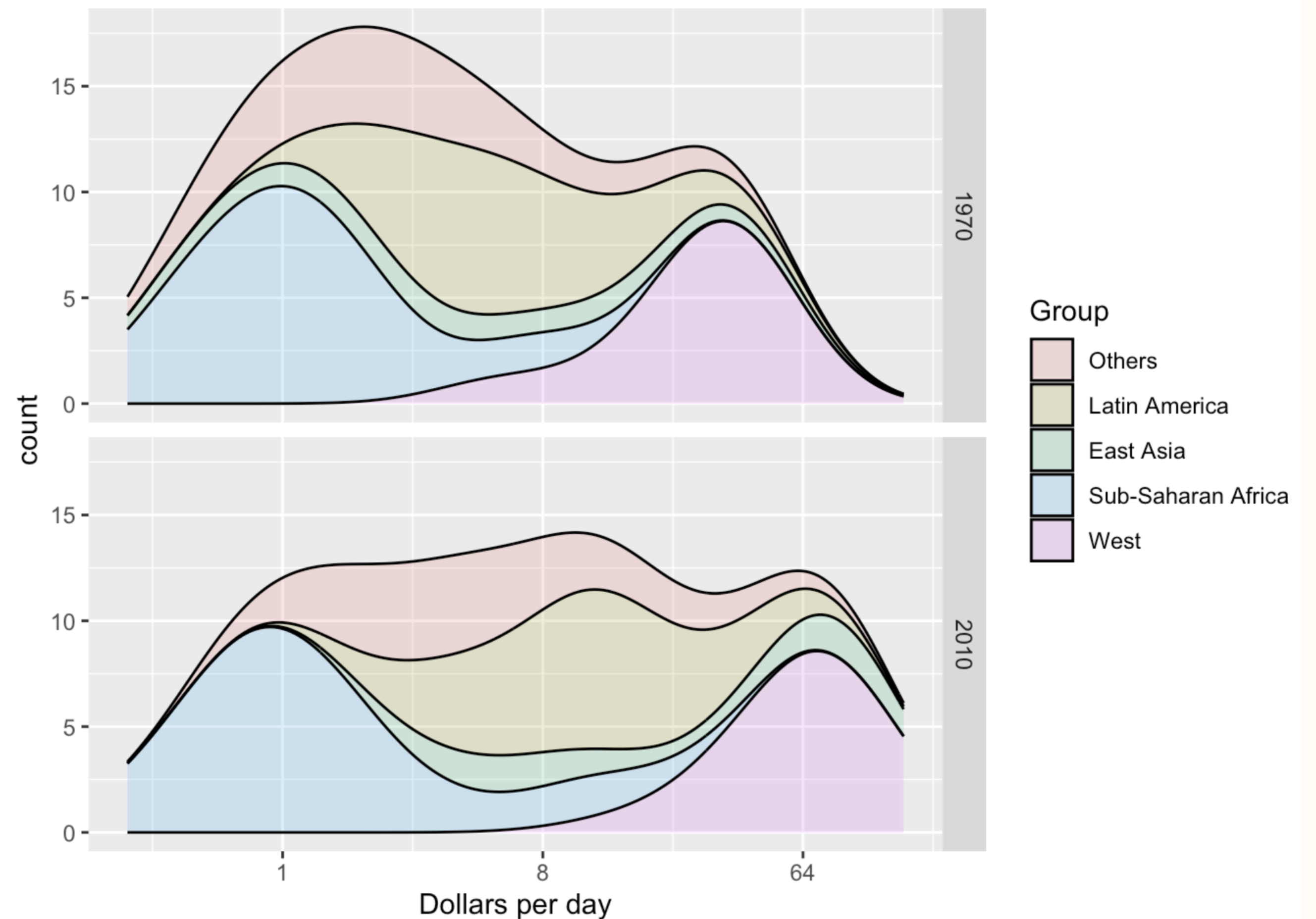
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



position = "stack"

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

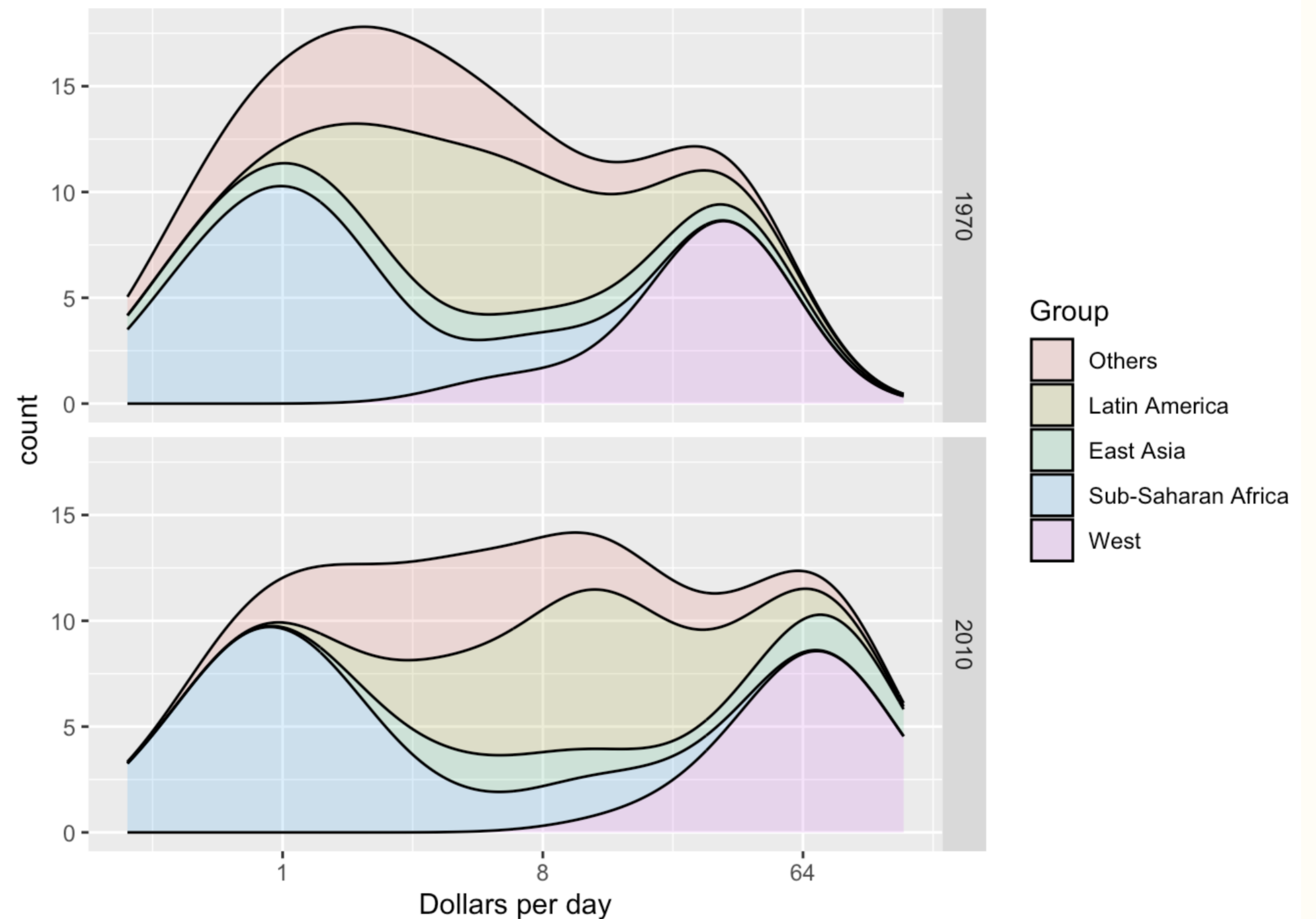
Plotting the two extremes (Sub-Saharan Africa and West) first shows the bimodality better



position = "stack"

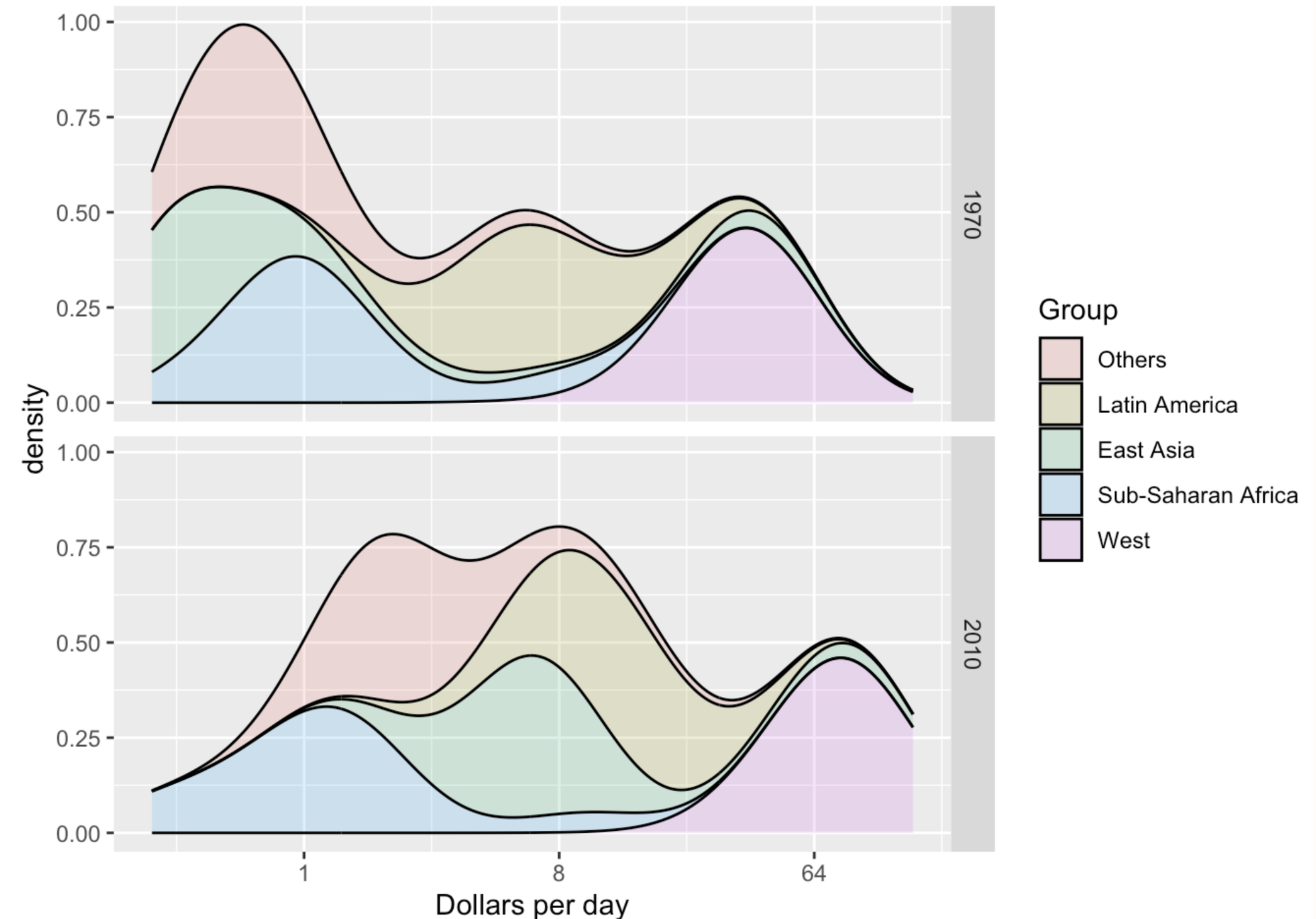
```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  ggplot(aes(dollars_per_day, y = after_stat(count),  
            fill = group)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```

Here, each country has the same weight, which can skew comparisons – it's better to weight countries according to population



Weighted densities (**weight**)

```
gapminder %>%  
  filter(year %in% c(past_year, present_year) &  
         country %in% country_list) %>%  
  group_by(year) %>%  
  mutate(weight = population/sum(population)) %>%  
  ungroup() %>%  
  ggplot(aes(dollars_per_day, fill = group, weight = weight)) +  
  scale_x_continuous(trans = "log2") +  
  geom_density(alpha = 0.2, bw = 0.75, position = "stack") +  
  facet_grid(year ~ .) +  
  xlab("Dollars per day") +  
  labs(fill = "Group")
```



Summary

- Distributions across groups can be compared using smooth density plots
- The **case_when** function is an efficient way to create a categorical (factor) variable
- Smooth density plots can be stacked using the **position = "stack"** argument
- Smooth density plots can be weighted using the **weight** argument

Maps

Heather Mattie, PhD



Maps Roadmap

1. The **maps** package
2. Plotting global life expectancy
3. Plotting US state murder rates
4. Case study: COVID-19 cases

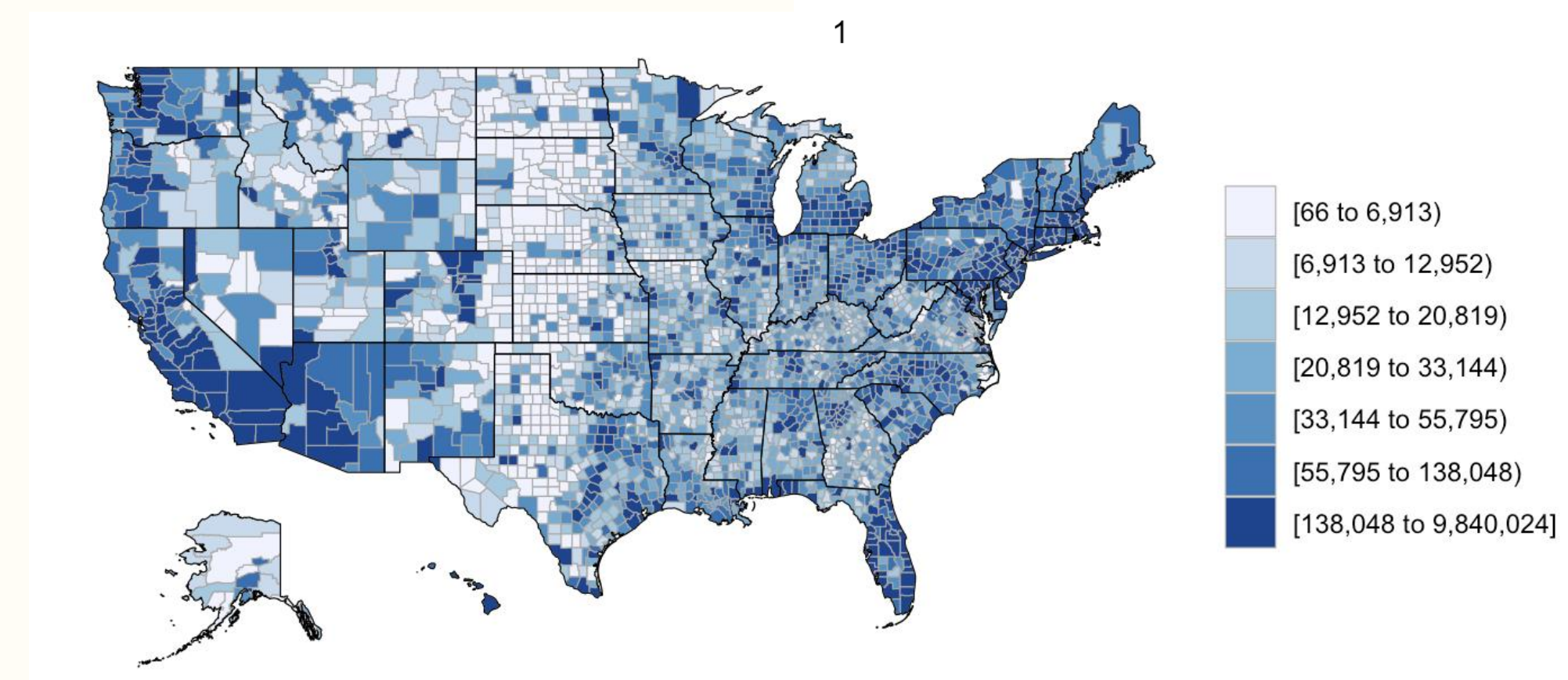
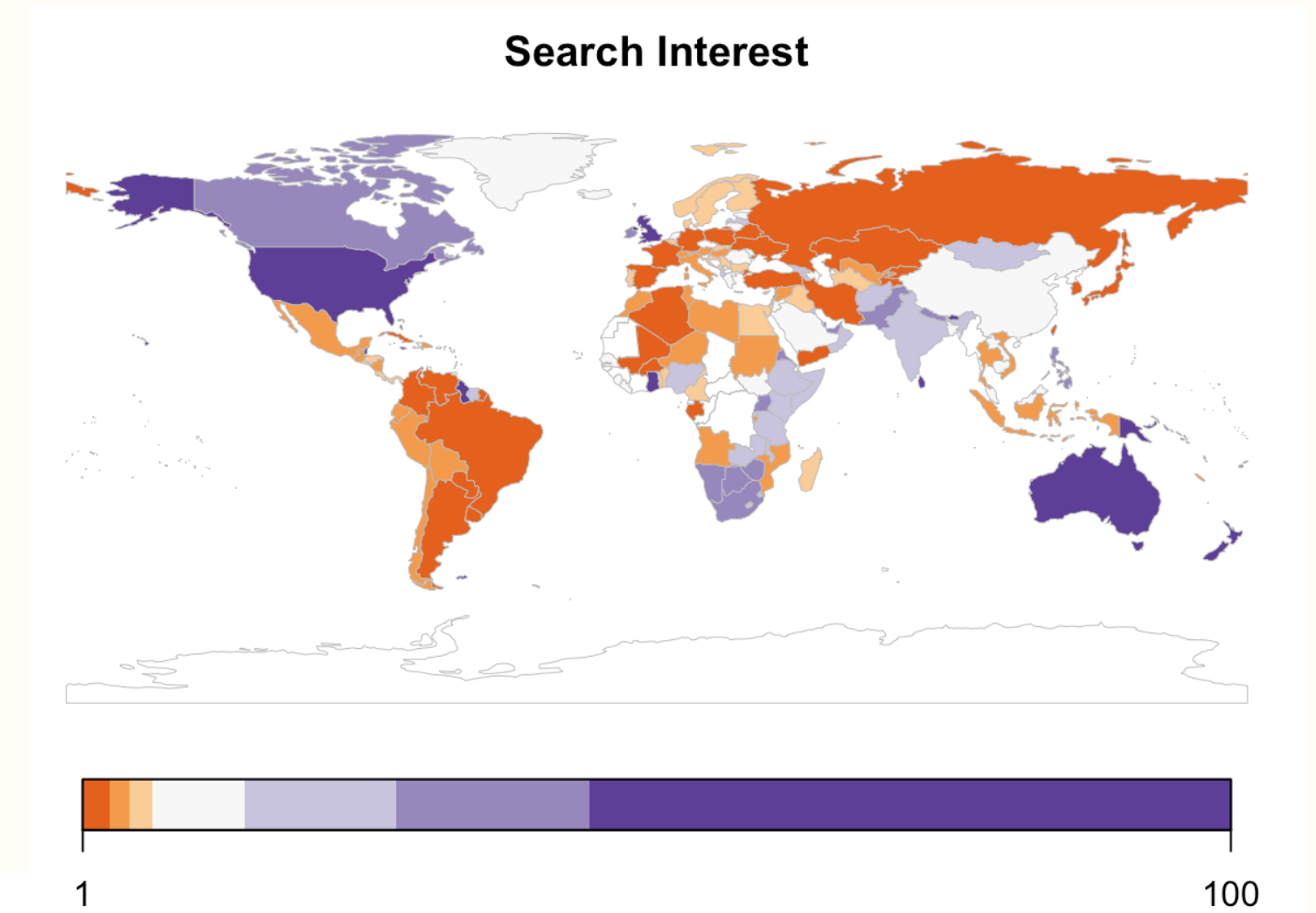
Maps

The **maps** package

Heather Mattie, PhD

Maps

- Plots of **maps** can be very powerful, informative and aesthetically pleasing
- They can also be misleading if not created correctly
- We will use the **maps** package to create plots of maps
- Other packages: **ggspatial**, **sf**, **urbnmapr**, **rnaturalearth**, etc.

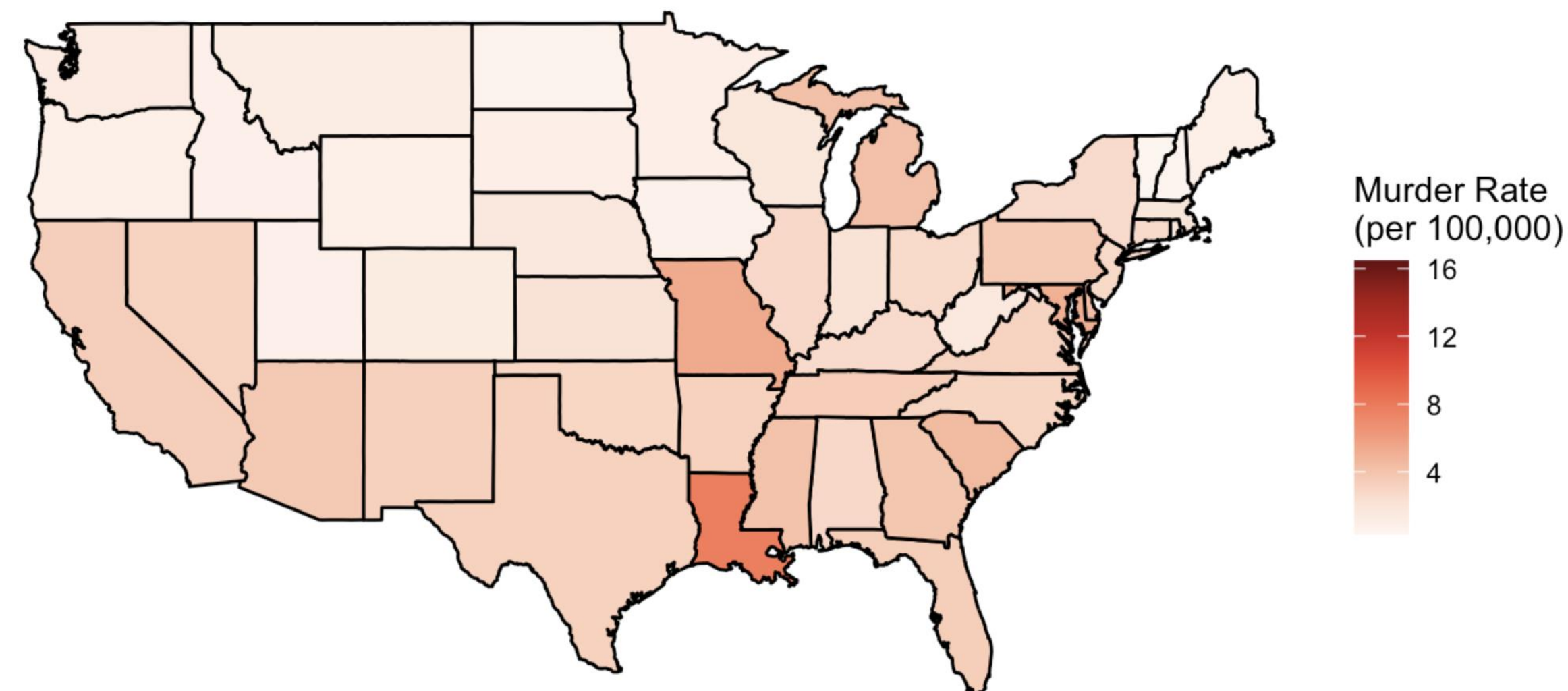


maps

The maps package contains outlines of continents, countries, states, and counties / jurisdictions / provinces

- Specify which map to use with `map_data()`
- Loaded as a data frame and can be used with the pipe (`%>%`) and `ggplot`

US Murder Rate by State



map_data("world")

- **long** = longitude
- **lat** = latitude
- **group** indicates which countries can be drawn together
- **order** indicates which order to connect the **lat** and **long** dots
- **region** = country
- **subregion** = county, political jurisdictions, etc.

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

##		long	lat	group	order	region	subregion
##	1	-69.89912	12.45200	1	1	Aruba	<NA>
##	2	-69.89571	12.42300	1	2	Aruba	<NA>
##	3	-69.94219	12.43853	1	3	Aruba	<NA>
##	4	-70.00415	12.50049	1	4	Aruba	<NA>
##	5	-70.06612	12.54697	1	5	Aruba	<NA>
##	6	-70.05088	12.59707	1	6	Aruba	<NA>

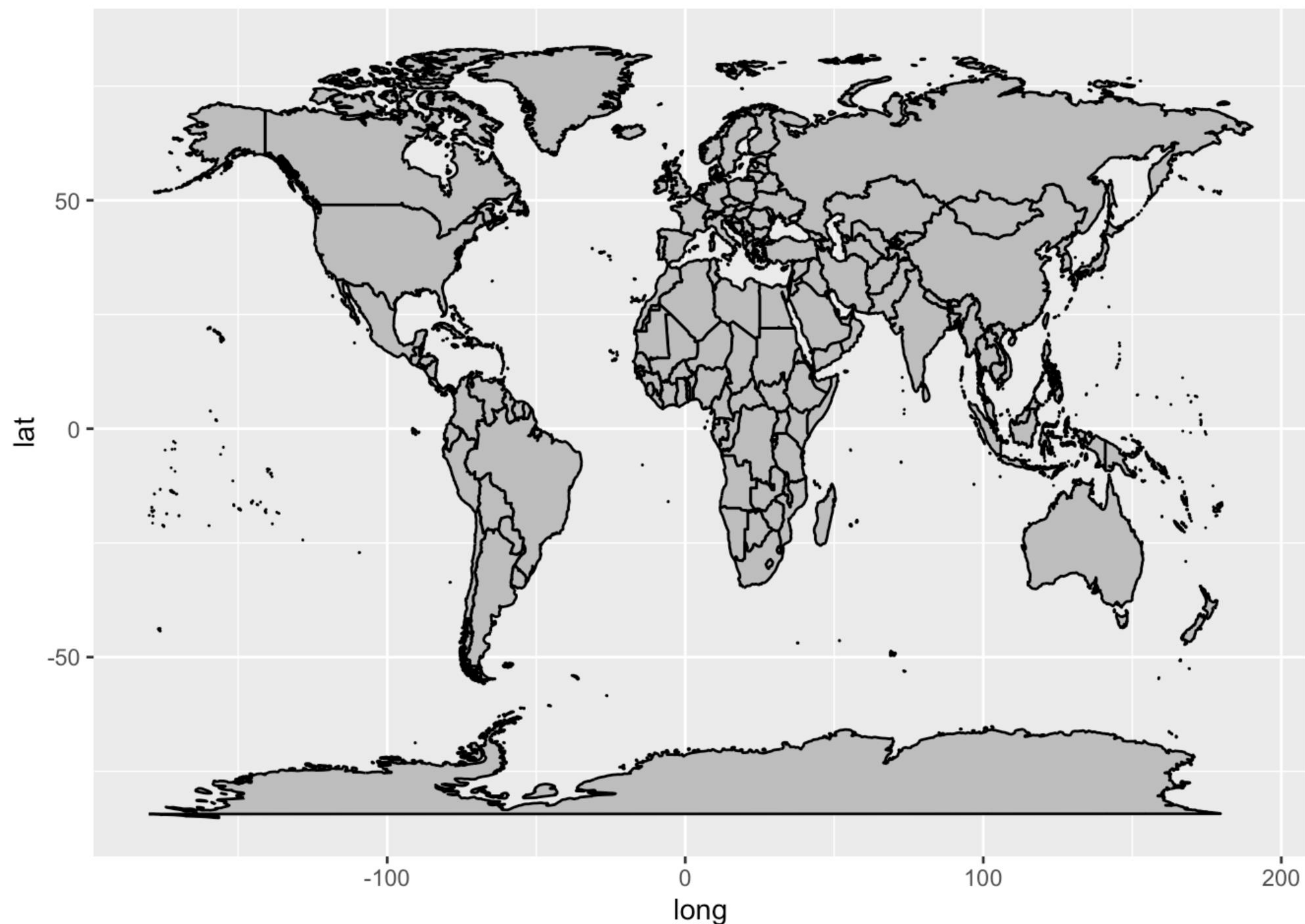
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```

map_data("world")

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##           long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba    <NA>
## 2 -69.89571 12.42300     1     2  Aruba    <NA>
## 3 -69.94219 12.43853     1     3  Aruba    <NA>
## 4 -70.00415 12.50049     1     4  Aruba    <NA>
## 5 -70.06612 12.54697     1     5  Aruba    <NA>
## 6 -70.05088 12.59707     1     6  Aruba    <NA>
```

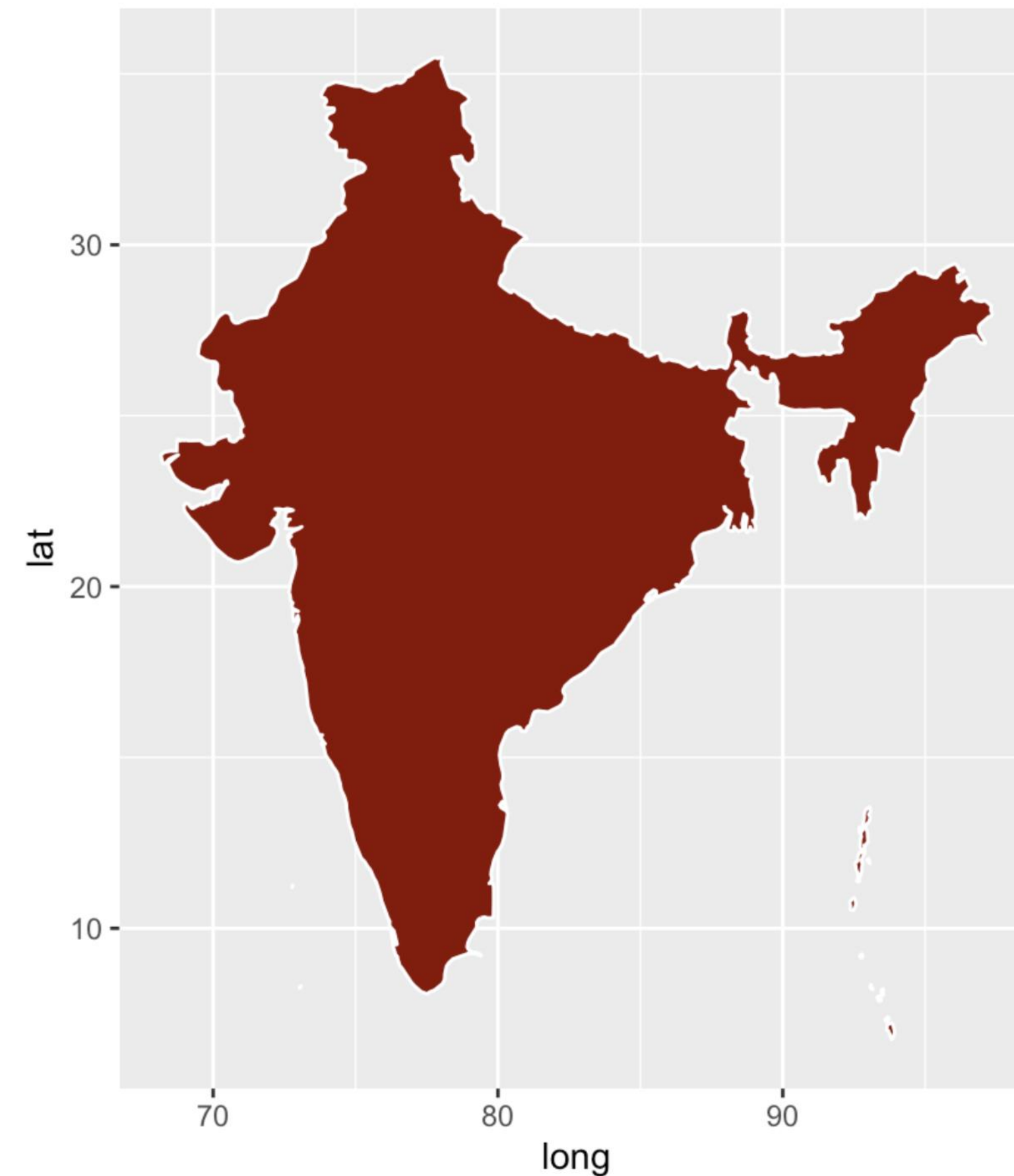
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```



Countries

```
india_map <- map_data("world", "India")

india_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.2)
```

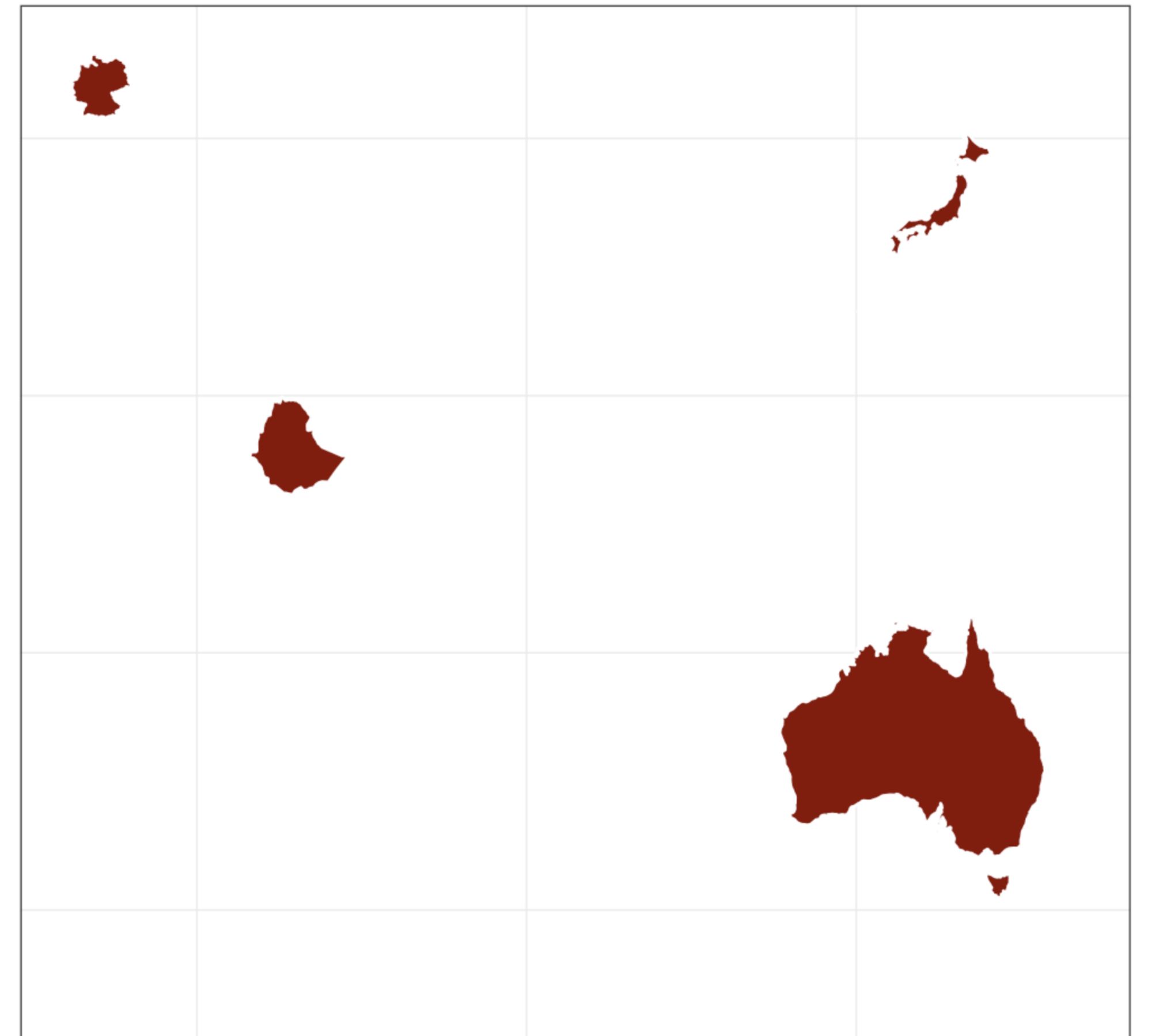


Countries

```
theme_set(theme_bw())

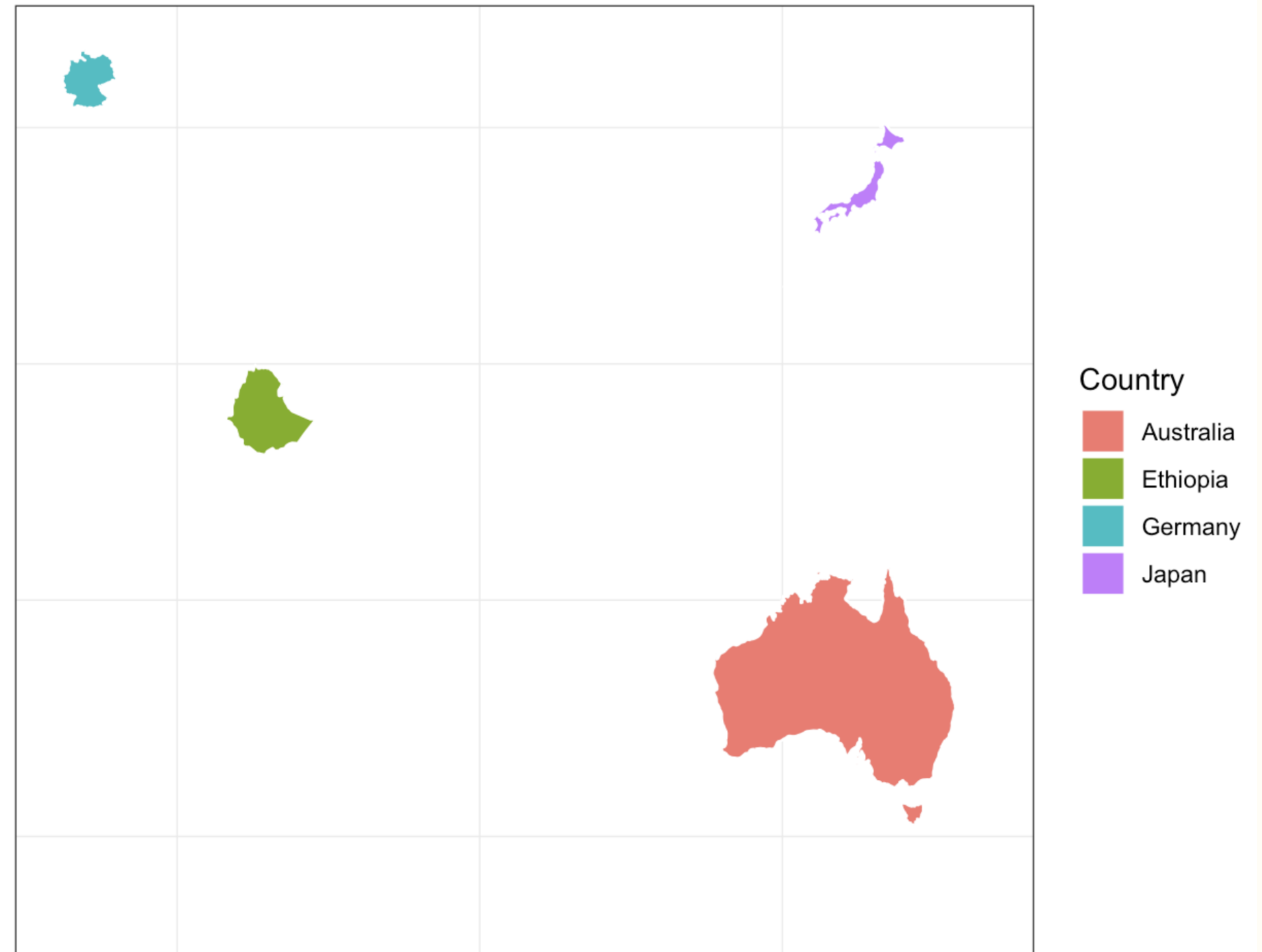
multiple_countries_map <- map_data("world",
                                   c("Japan", "Ethiopia", "Australia", "Germany"))

multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3)
```



Countries

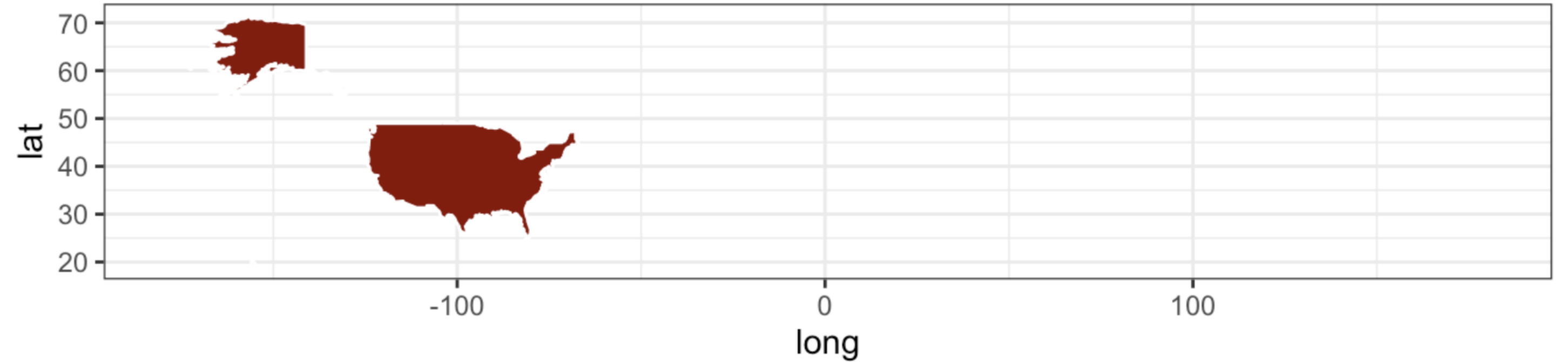
```
multiple_countries_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = region)) +  
  geom_polygon(color = "white") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  labs(fill = "Country") +  
  coord_fixed(1.3)
```



US maps

```
us_map <- map_data("world", "usa")

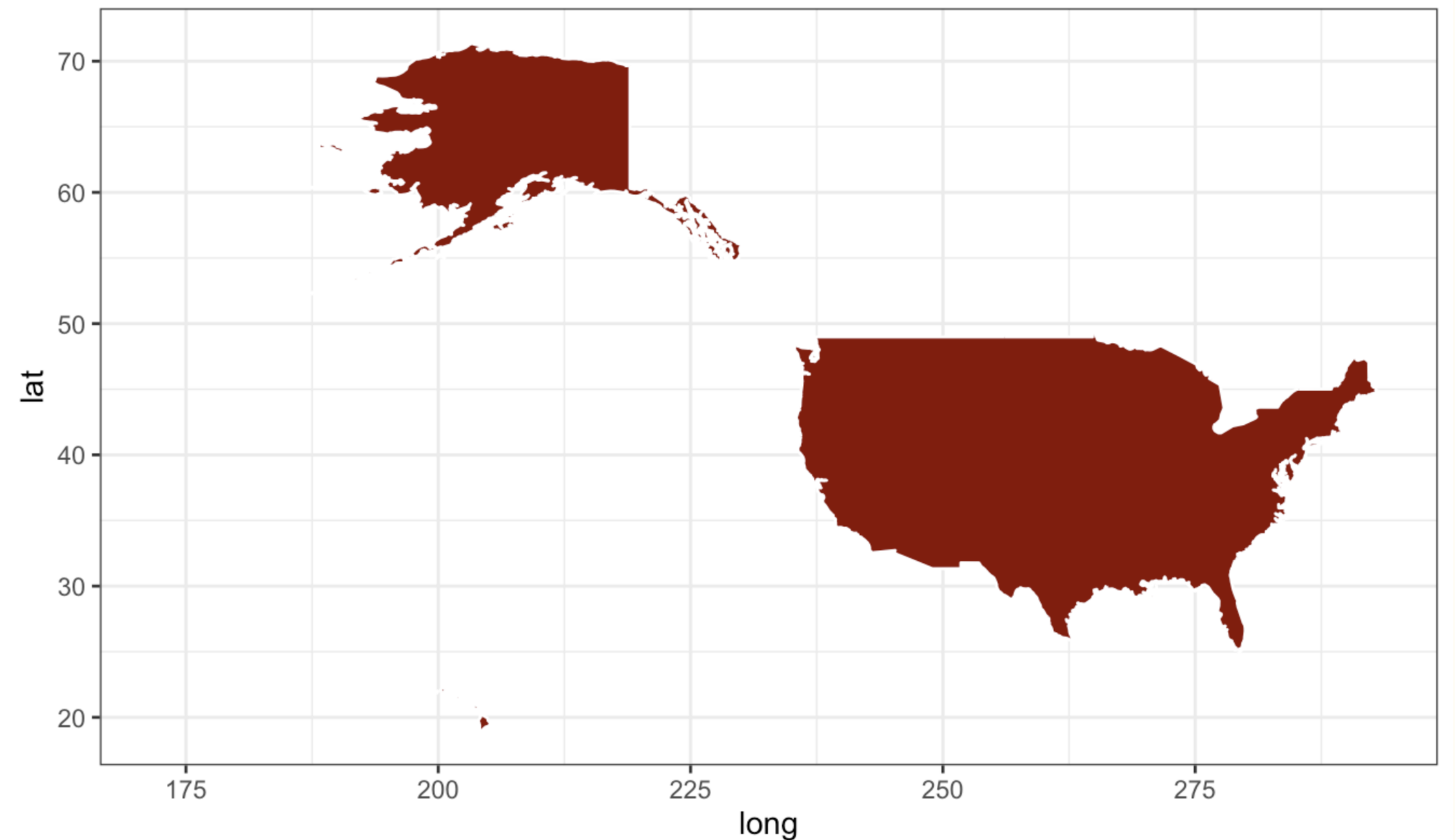
us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
better_us_map <- map_data("world2", "USA")

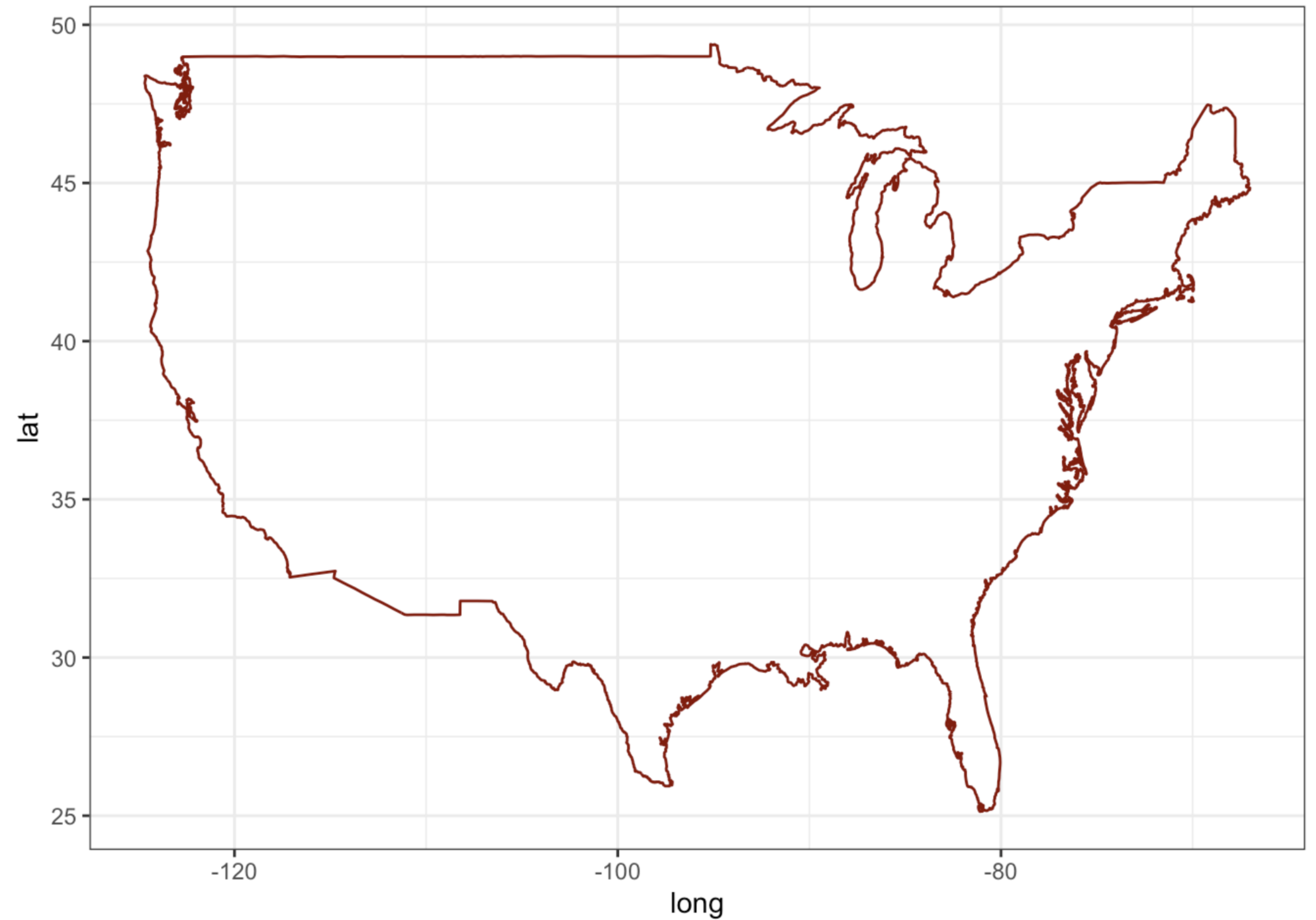
better_us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
usa <- map_data("usa")

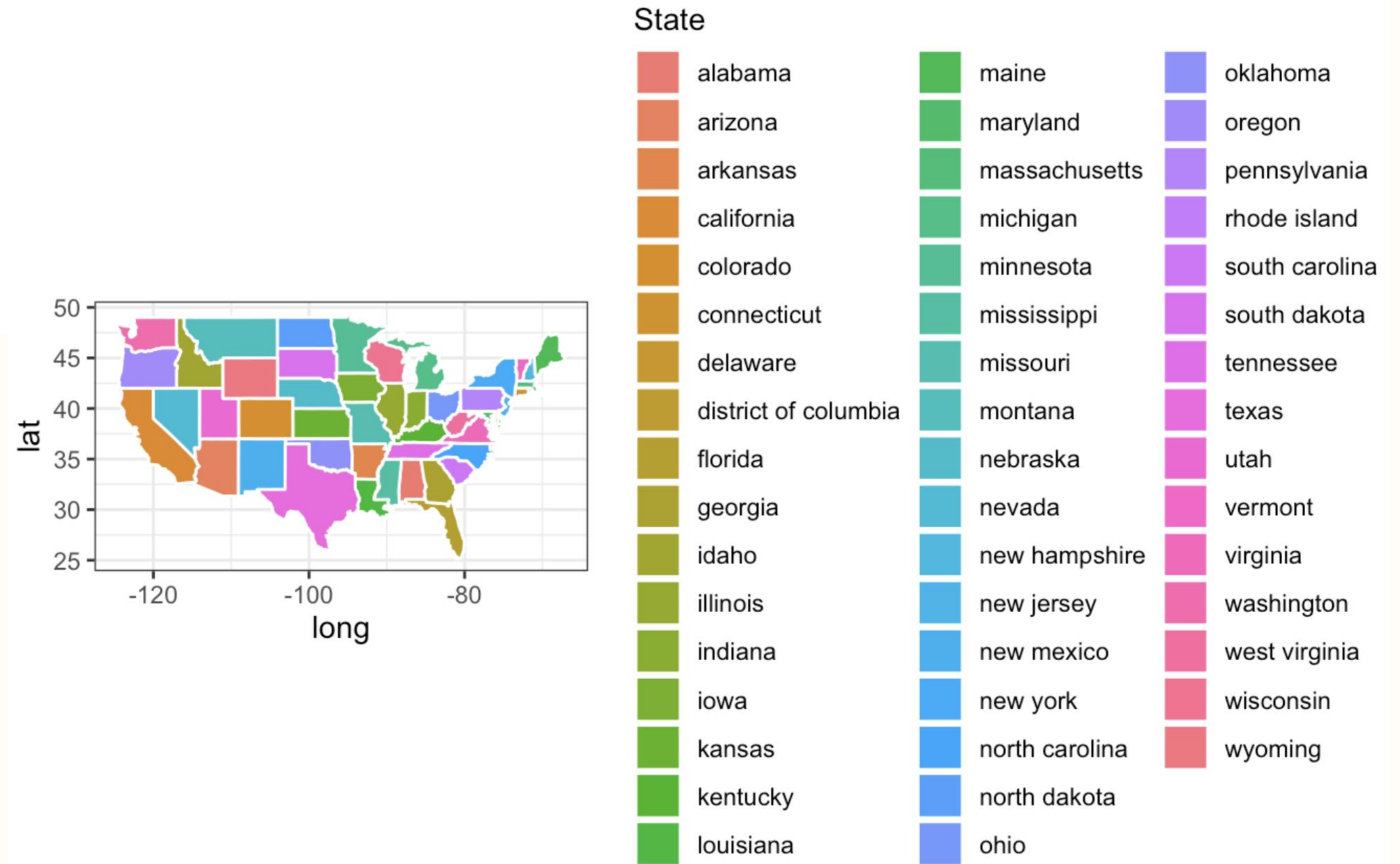
usa %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



US states

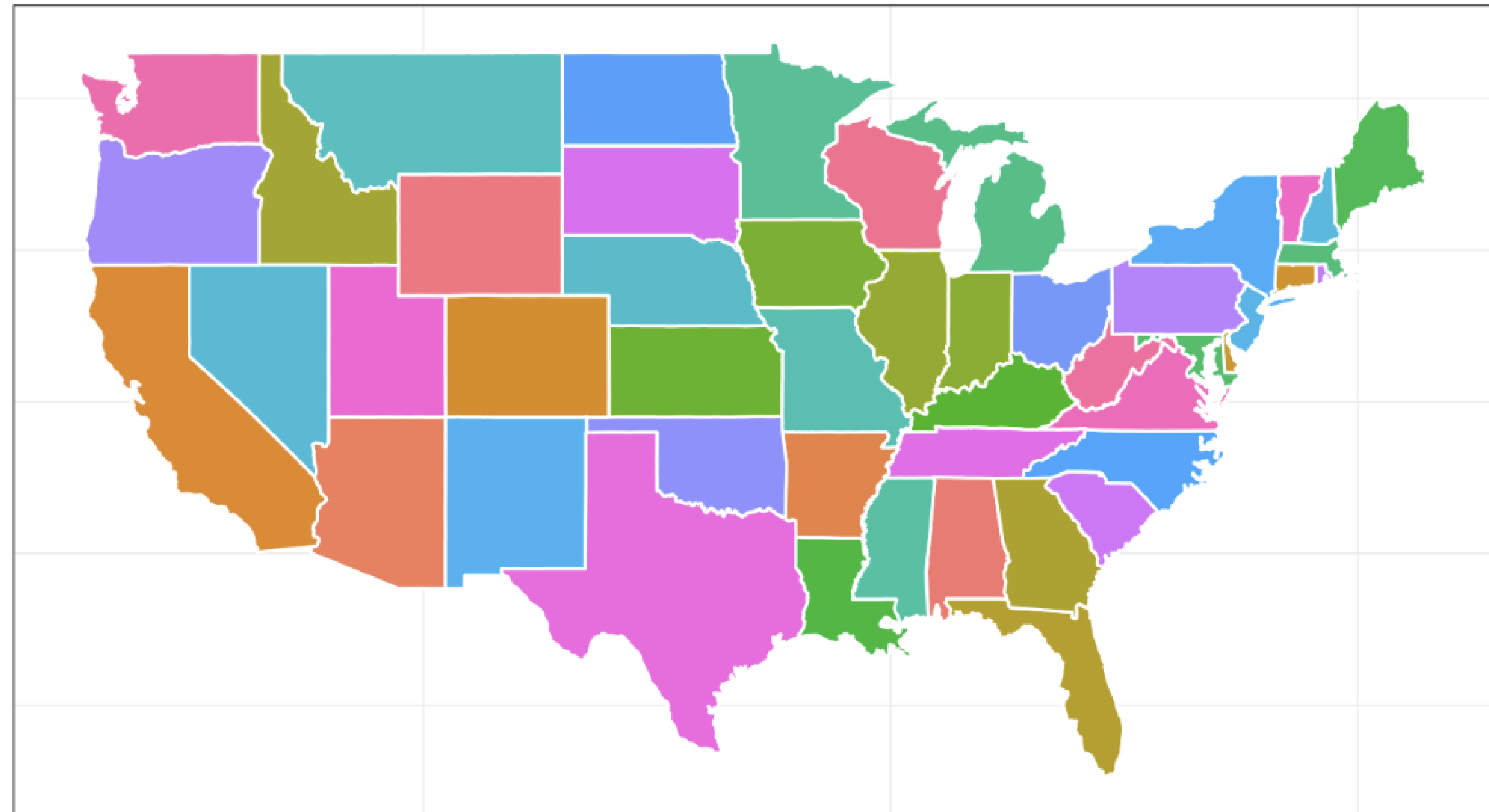
```
us_states <- map_data("state")

us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  labs(fill = "State") +
  coord_fixed(1.3)
```



US states

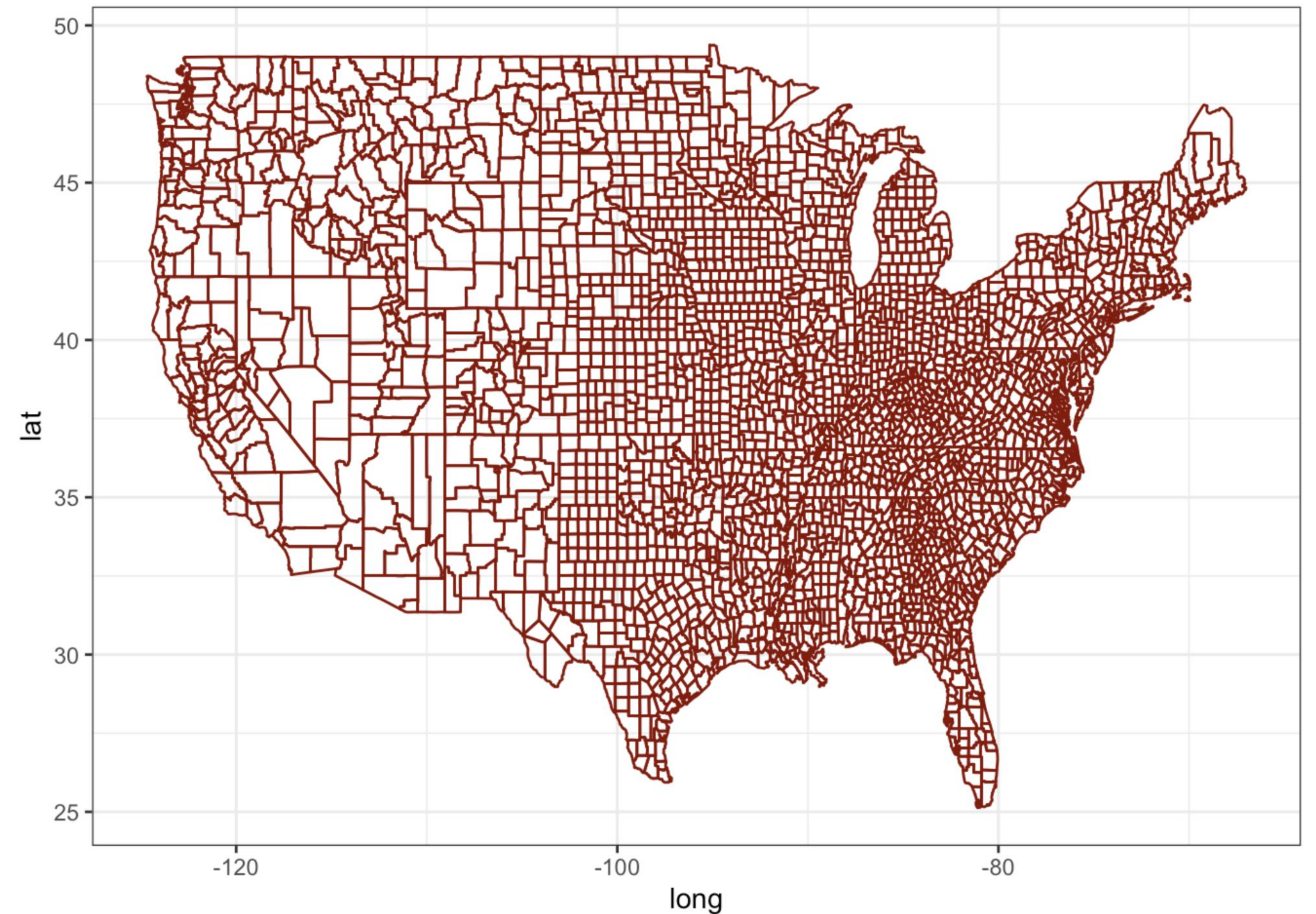
```
us_states %>%  
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +  
  geom_polygon(color = "white") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  guides(fill = "none")
```



US counties

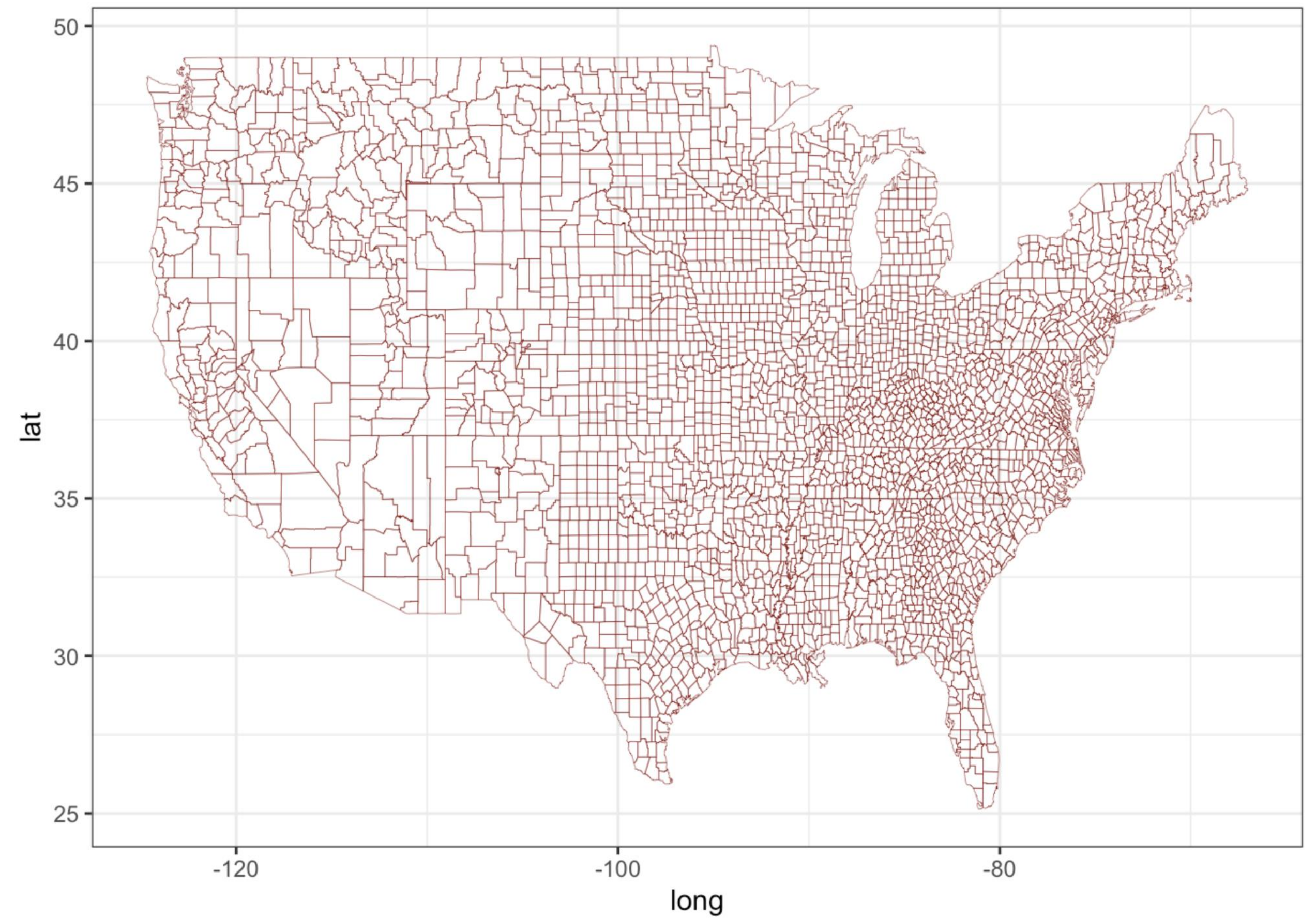
```
AllCounty <- map_data("county")
```

```
AllCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(color = "darkred", fill = NA)
```



US counties

```
AllCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(color = "darkred", fill = NA, linewidth = .1)
```



Summary

- The **maps** package can be used to produce maps of continents, countries, states, counties, and provinces
- The **map_data()** function loads map data
- Color or labels should be used instead of a legend if there are too many categories (countries, states, etc.) represented in the map

Maps

Plotting global life expectancy

Heather Mattie, PhD

Plotting global life expectancy

Task: plot a map displaying the life expectancy in 2015 for countries with data in the **gapminder** dataset

```
library(plyr)
library(tidyverse)
library(maps)
library(dslabs)

data(gapminder)
gapminder_2015 <- gapminder %>%
  filter(year == 2015)

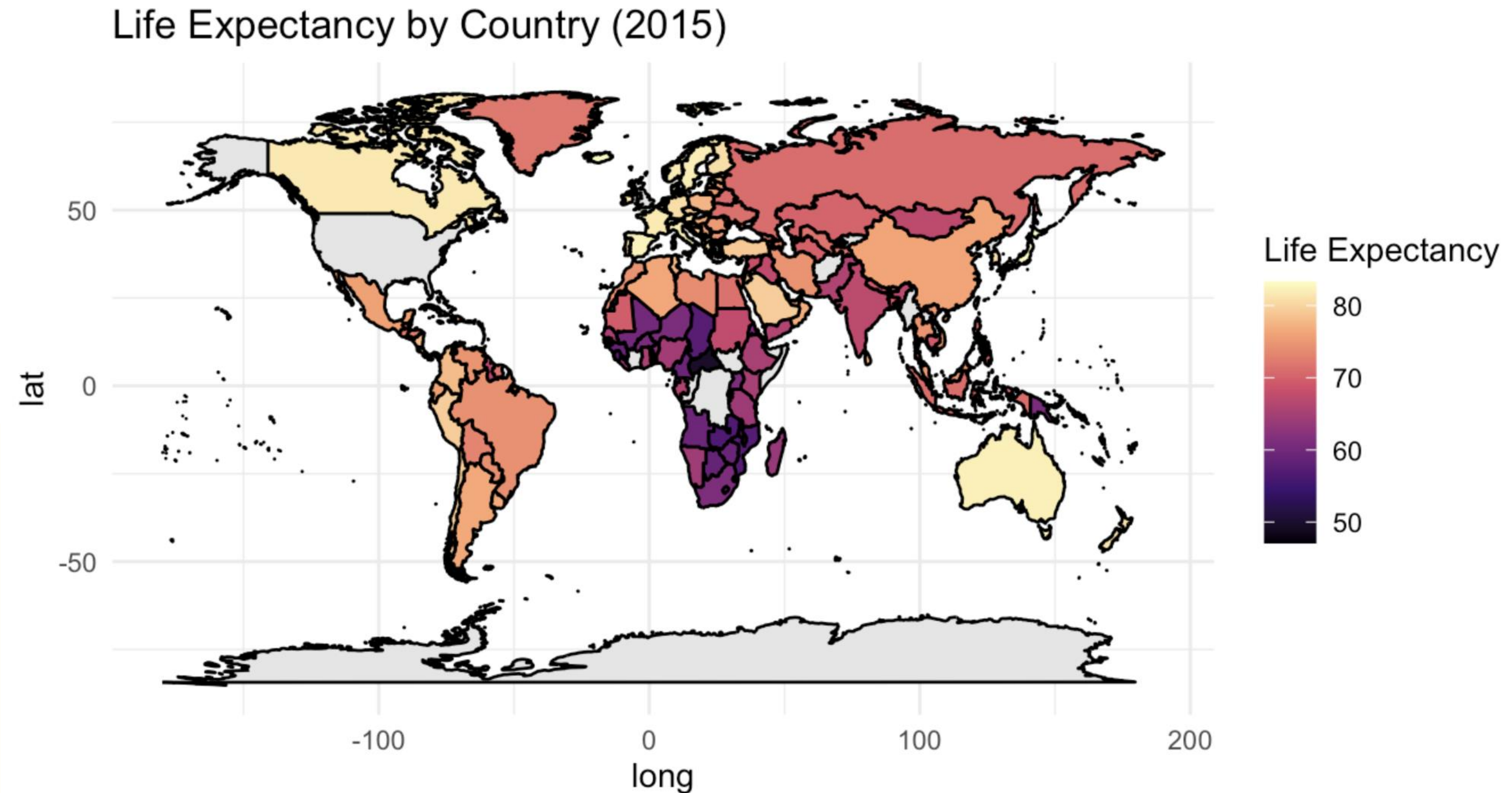
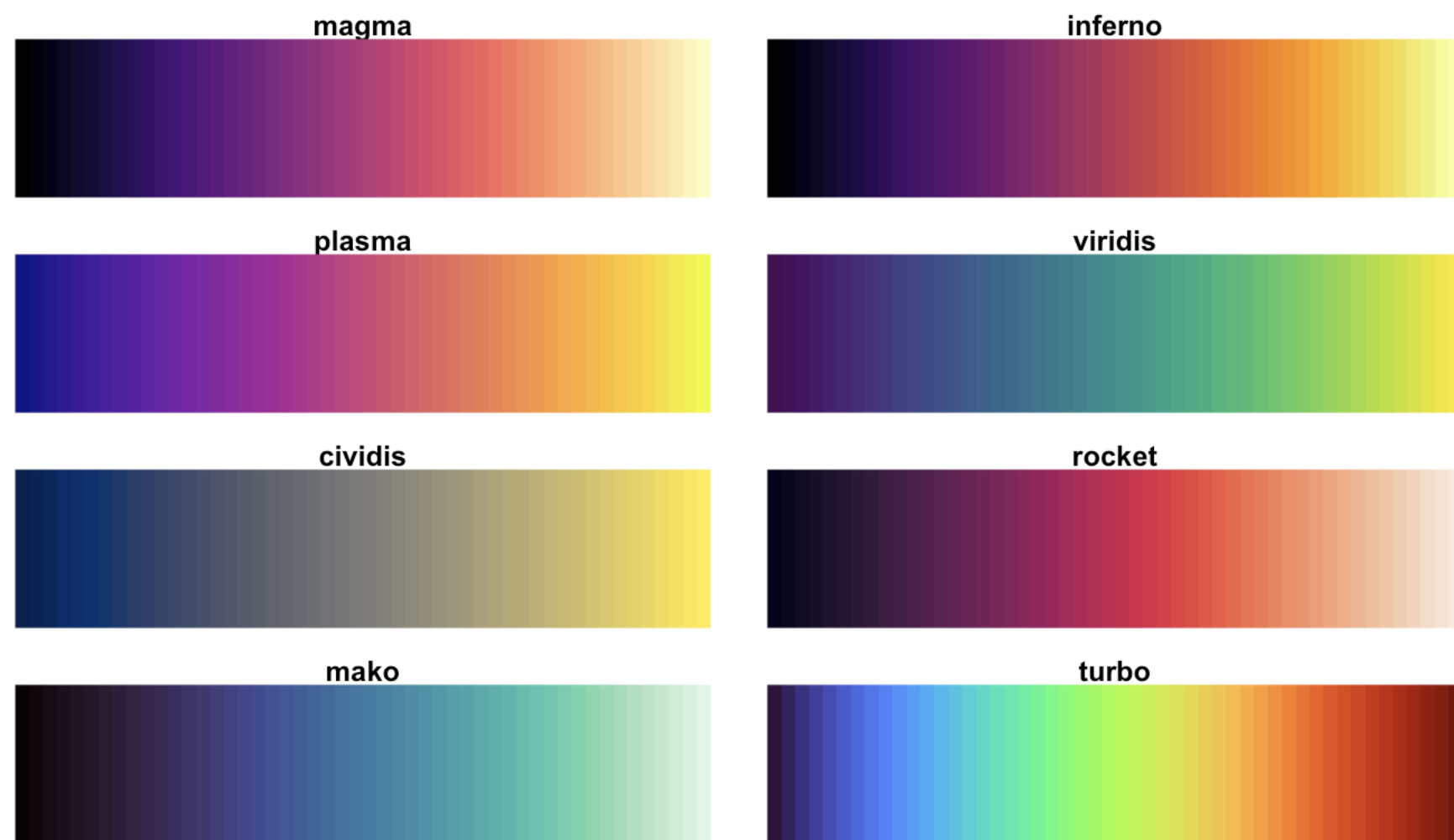
world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015,
                          by = c("region" = "country"))

head(merged_world)
```

```
##      long      lat group order region subregion year infant_mortality
## 1 -69.89912 12.45200     1     1  Aruba      <NA> 2015              NA
## 2 -69.89571 12.42300     1     2  Aruba      <NA> 2015              NA
## 3 -69.94219 12.43853     1     3  Aruba      <NA> 2015              NA
## 4 -70.00415 12.50049     1     4  Aruba      <NA> 2015              NA
## 5 -70.06612 12.54697     1     5  Aruba      <NA> 2015              NA
## 6 -70.05088 12.59707     1     6  Aruba      <NA> 2015              NA
##      life_expectancy fertility population gdp continent region.y
## 1             75.72      1.66     103889   NA  Americas Caribbean
## 2             75.72      1.66     103889   NA  Americas Caribbean
## 3             75.72      1.66     103889   NA  Americas Caribbean
## 4             75.72      1.66     103889   NA  Americas Caribbean
## 5             75.72      1.66     103889   NA  Americas Caribbean
## 6             75.72      1.66     103889   NA  Americas Caribbean
```

Plotting global life expectancy

```
ggplot(merged_world, aes(long, lat, group = group,  
                          fill = life_expectancy)) +  
  geom_polygon(color = "black") +  
  scale_fill_viridis_c(option = "magma",  
                       na.value = "gray90") +  
  theme_minimal() +  
  labs(title = "Life Expectancy by Country (2015)",  
       fill = "Life Expectancy") +  
  coord_fixed(1.3)
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
## [5] "Antarctica"       "Antigua"
## [7] "Ascension Island" "Azores"
## [9] "Barbuda"          "Bermuda"
## [11] "Bonaire"          "Canary Islands"
## [13] "Cayman Islands"   "Chagos Archipelago"
## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"      "Jersey"
## [33] "Kosovo"           "Kyrgyzstan"
## [35] "Laos"             "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"          "Algeria"
## [3] "Angola"           "Antigua and Barbuda"
## [5] "Argentina"        "Armenia"
## [7] "Aruba"            "Australia"
## [9] "Austria"          "Azerbaijan"
## [11] "Bahamas"          "Bahrain"
## [13] "Bangladesh"       "Barbados"
## [15] "Belarus"          "Belgium"
## [17] "Belize"           "Benin"
## [19] "Bhutan"           "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"           "Brunei"
## [25] "Bulgaria"         "Burkina Faso"
## [27] "Burundi"         "Cambodia"
## [29] "Cameroon"         "Canada"
## [31] "Cape Verde"       "Central African Republic"
## [33] "Chad"             "Chile"
## [35] "China"            "Colombia"
## [37] "Comoros"          "Congo, Dem. Rep."
## [39] "Congo, Rep."      "Costa Rica"
## [41] "Cote d'Ivoire"    "Croatia"
## [43] "Cuba"             "Cyprus"
## [45] "Czech Republic"  "Denmark"
## [47] "Djibouti"         "Dominican Republic"
## [49] "Ecuador"          "Egypt"
## [51] "El Salvador"      "Equatorial Guinea"
## [53] "Eritrea"          "Estonia"
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

## [1] "Afghanistan"	"American Samoa"
## [3] "Andorra"	"Anguilla"
## [5] "Antarctica"	"Antigua"
## [7] "Ascension Island"	"Azores"
## [9] "Barbuda"	"Bermuda"
## [11] "Bonaire"	"Canary Islands"
## [13] "Cayman Islands"	"Chagos Archipelago"
## [15] "Christmas Island"	"Cocos Islands"
## [17] "Cook Islands"	"Curacao"
## [19] "Democratic Republic of the Congo"	"Dominica"
## [21] "Falkland Islands"	"Faroe Islands"
## [23] "French Guiana"	"French Southern and Antarctic Lands"
## [25] "Grenadines"	"Guadeloupe"
## [27] "Guam"	"Guernsey"
## [29] "Heard Island"	"Isle of Man"
## [31] "Ivory Coast"	"Jersey"
## [33] "Kosovo"	"Kyrgyzstan"
## [35] "Laos"	"Liechtenstein"
## [37] "Madeira Islands"	"Marshall Islands"
## [39] "Martinique"	"Mayotte"
## [41] "Micronesia"	"Monaco"
## [43] "Montserrat"	"Myanmar"
## [45] "Nauru"	"Nevis"

```
levels(gapminder_2015$country)
```

## [1] "Albania"	"Algeria"
## [3] "Angola"	"Antigua and Barbuda"
## [5] "Argentina"	"Armenia"
## [7] "Aruba"	"Australia"
## [9] "Austria"	"Azerbaijan"
## [11] "Bahamas"	"Bahrain"
## [13] "Bangladesh"	"Barbados"
## [15] "Belarus"	"Belgium"
## [17] "Belize"	"Benin"
## [19] "Bhutan"	"Bolivia"
## [21] "Bosnia and Herzegovina"	"Botswana"
## [23] "Brazil"	"Brunei"
## [25] "Bulgaria"	"Burkina Faso"
## [27] "Burundi"	"Cambodia"
## [29] "Cameroon"	"Canada"
## [31] "Cape Verde"	"Central African Republic"
## [33] "Chad"	"Chile"
## [35] "China"	"Colombia"
## [37] "Comoros"	"Congo, Dem. Rep."
## [39] "Congo, Rep."	"Costa Rica"
## [41] "Cote d'Ivoire"	"Croatia"
## [43] "Cuba"	"Cyprus"
## [45] "Czech Republic"	"Denmark"
## [47] "Djibouti"	"Dominican Republic"
## [49] "Ecuador"	"Egypt"
## [51] "El Salvador"	"Equatorial Guinea"
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Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
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## [9] "Barbuda"          "Bermuda"
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## [13] "Cayman Islands"   "Chagos Archipelago"
## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"      "Jersey"
## [33] "Kosovo"           "Kyrgyzstan"
## [35] "Laos"             "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"      "Algeria"
## [3] "Angola"       "Antigua and Barbuda"
## [5] "Argentina"    "Armenia"
## [7] "Aruba"        "Australia"
## [9] "Austria"      "Azerbaijan"
## [11] "Bahamas"      "Bahrain"
## [13] "Bangladesh"   "Barbados"
## [15] "Belarus"      "Belgium"
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## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"       "Brunei"
## [25] "Bulgaria"     "Burkina Faso"
## [27] "Burundi"     "Cambodia"
## [29] "Cameroon"     "Canada"
## [31] "Cape Verde"   "Central African Republic"
## [33] "Chad"         "Chile"
## [35] "China"        "Colombia"
## [37] "Comoros"      "Congo, Dem. Rep."
## [39] "Congo, Rep."  "Costa Rica"
## [41] "Cote d'Ivoire" "Croatia"
## [43] "Cuba"         "Cyprus"
## [45] "Czech Republic" "Denmark"
## [47] "Djibouti"     "Dominican Republic"
## [49] "Ecuador"      "Egypt"
## [51] "El Salvador"  "Equatorial Guinea"
## [53] "Eritrea"      "Estonia"
```

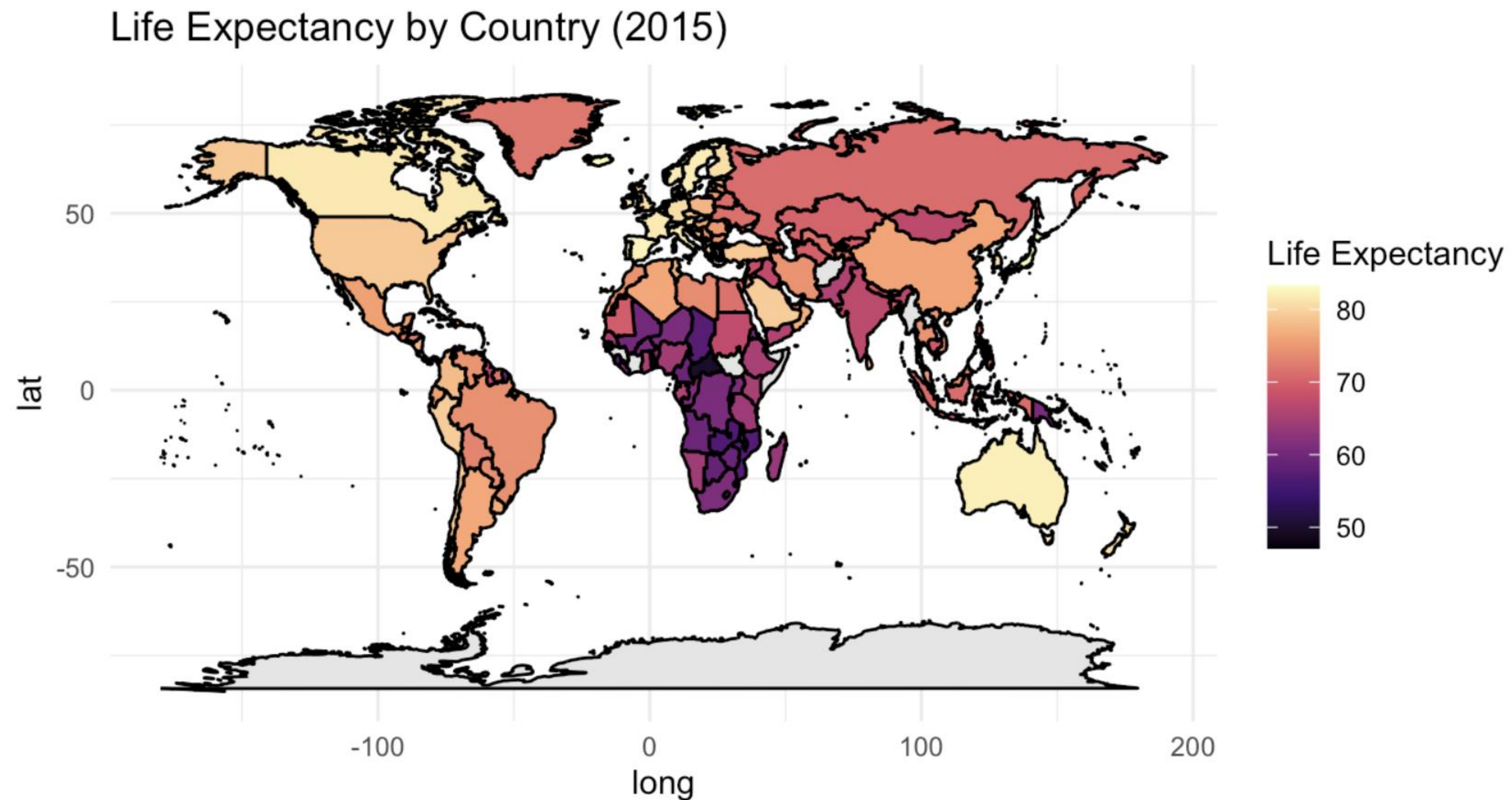


Renaming

```
gapminder_2015 <- gapminder %>% filter(year == 2015) %>%  
  mutate(country =  
    revalue(country, c("Antigua and Barbuda" = "Antigua",  
                        "Samoa" = "American Samoa",  
                        "Congo, Rep." = "Republic of Congo",  
                        "Congo, Dem. Rep." = "Democratic Republic of the Congo",  
                        "Guinea" = "French Guiana",  
                        "Kyrgyz Republic" = "Kyrgyzstan",  
                        "St. Lucia" = "Saint Lucia",  
                        "St. Vincent and the Grenadines" = "Saint Vincent",  
                        "Trinidad and Tobago" = "Trinidad",  
                        "United Kingdom" = "UK",  
                        "United States" = "USA"))))
```

```
world_map <- map_data("world")  
merged_world <- left_join(world_map, gapminder_2015, by = c("region" = "country"))
```

```
ggplot(merged_world, aes(long, lat, group = group, fill = life_expectancy)) +  
  geom_polygon(color = "black") +  
  scale_fill_viridis_c(option = "magma", na.value = "gray90") +  
  theme_minimal() +  
  labs(title = "Life Expectancy by Country (2015)", fill = "Life Expectancy") +  
  coord_fixed(1.3)
```



Summary

- Inconsistencies in the spelling of locations can lead to missing data in plots of maps
- Renaming the locations in one data frame before joining to a second data frame can solve a lot of inconsistencies
- Sometimes new data will need to be collected to complete a map plot